

The Uncommon Sense Effect: experimental evidence for underspecification as the mental basis of logical inference

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Abstract

Logical reasoning is one of humanity’s most powerful abilities. A widespread assumption across psychology, linguistics, and philosophy holds that reasoning operates over concepts that refer to objects and properties in the world, yet this has rarely been tested empirically. We introduce a novel paradigm that exploits lexical ambiguity to differentiate candidate representations for human inference: word-forms, reference-fixing concepts, or more abstract “underspecified representations” that constrain meaning without fully determining it. Across three experiments ($N = 158$), participants evaluated deductive arguments equivocating over polysemous or homonymous terms. Although these arguments are logically equivalent under standard analyses, participants reliably accepted non-truth-preserving polysemous arguments while rejecting homonymous ones. This asymmetry—the Uncommon Sense Effect—points to underspecified representations as the default basis of logical inference. This challenges a foundational assumption about meaning, and opens new empirical avenues for studying the link between what we think and why we think it.

Logical inference is ubiquitous in human experience, manifesting early and operating throughout our lives. Preverbal infants deploy inferential reasoning to track the locations of physical objects¹ and to interpret others’ social behavior². Young children rely on logical inference to learn the meanings of words^{3–5}. In adulthood, even without formal instruction, people rapidly, automatically, and unconsciously apply inference patterns, including rules corresponding to classical schemas such as *modus ponens*^{6–9}. The ability to reason in an abstract, symbolic system may be a core part of what makes us human^{10–12}.

Part of what makes human inferential abilities so powerful is their topic-neutrality, allowing us to reason about an indefinite number of ideas. This topic-neutrality suggests that logical operations are sensitive primarily to the *structure* of thoughts, rather than their *content*^{13–16}. For example, inferring from the statements “Some whales weigh over 150 tons” and “All whales are mammals” to the conclusion “Some mammals weigh over 150 tons” does not depend on any special properties, or even knowledge, of whales, mammals, or weight. Rather, this operation

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merely relies on the argument fulfilling a certain structural schema: “Some M are P,” “All M are S,” therefore “Some S are P.” It is an open empirical question which inferential schemas humans employ, and whether any are psychologically basic^{1;17–23}. Here, however, we are concerned not with identifying the specific logical rules people use, but with a more general question: to which kinds of mental representations do these rules apply? More succinctly, when we infer, what are we inferring over?

The answer tacitly favored across psychology, linguistics, and philosophy is that the mental structures over which we reason and compute meaning are the same representations that can be true or false^{24–27}. Across cognitive science, meaning is defined denotationally, in terms of extensions — via functions mapping possible worlds to truth values, individuals, or other entities^{25;28–30} — with entailment relations as a key data point. This assumption is so foundational and widespread that it is rarely treated as a substantive empirical or theoretical claim in need of defense.

Under this denotational assumption, the representations that figure in inference are those that pick out determinate objects in the world. For example, the concept *WHALES* fixes a denotation by representing the (possible) set of whales, and it is this denotation-determining representation that is taken to enter into inferential processes. Theories of concepts differ over how this reference relationship is achieved — whether via a causal history^{31;32}, an informational relationship^{33;34}, or some other mechanism³⁵. Nevertheless, all these approaches share the assumption that concepts are both the bearers of reference and the mental units over which inference rules apply. This identification has an intuitive appeal: if inferential schemas operate over denotations, then valid schemas are guaranteed to preserve truth.

Despite its prominence, the denotational assumption has rarely been subjected to direct empirical scrutiny. Laboratory experiments on human inference typically task participants with evaluating arguments constructed from *unambiguous* sentences¹⁷. While such studies may yield insight into which inferential schemas people do or do not possess, they cannot uncover the kinds of mental representation over which those schemas are processed. By controlling for ambiguity, previous paradigms introduce a critical confound: lexically unambiguous words share many levels of representational structure. Returning to the earlier example, processing the two premises “Some whales weigh over 150 tons” and “All whales are mammals” may involve two tokens of a representation referring to the set of whales (*Cetacea*). But it also involves recognizing two instances of the same orthographic string, “whales.” Moreover, growing evidence suggests that words in the mind additionally involve *underspecified* representations that constrain, without fully determining, their denotations^{36–40}. As a result, existing paradigms cannot adjudicate between these representational levels. This leaves open the question of whether human inference operates over orthographic forms, reference-determining denotations, or underspecified representations.

Here, we introduce a novel paradigm that tests human inference over lexically ambiguous words — words that share some, but not all, levels of representational structure (see Figure 1). Lexical ambiguity therefore provides a principled way to dissociate orthographic, denotational, and underspecified representations within otherwise valid deductive arguments. One well-studied source of lexical ambiguity is homonymy: items which share an orthographic form (e.g., “bat”) but correspond to multiple, unrelated denotations (e.g., flying mammals or baseball equipment). Homonymy is traditionally contrasted with polysemy^{41;42}, in which a single word form maps to multiple, related senses. For instance, “dinner” can refer to both an evening dining event and to the food consumed at that event. Although related, these senses are nevertheless denotationally distinct — foods can be overcooked while events cannot. Crucially, psycholinguistic evidence suggests that, unlike homonyms, the mental representation of polysemes involves a shared, underspecified representation that is neutral between the word’s denotation-determining senses^{43–47}.

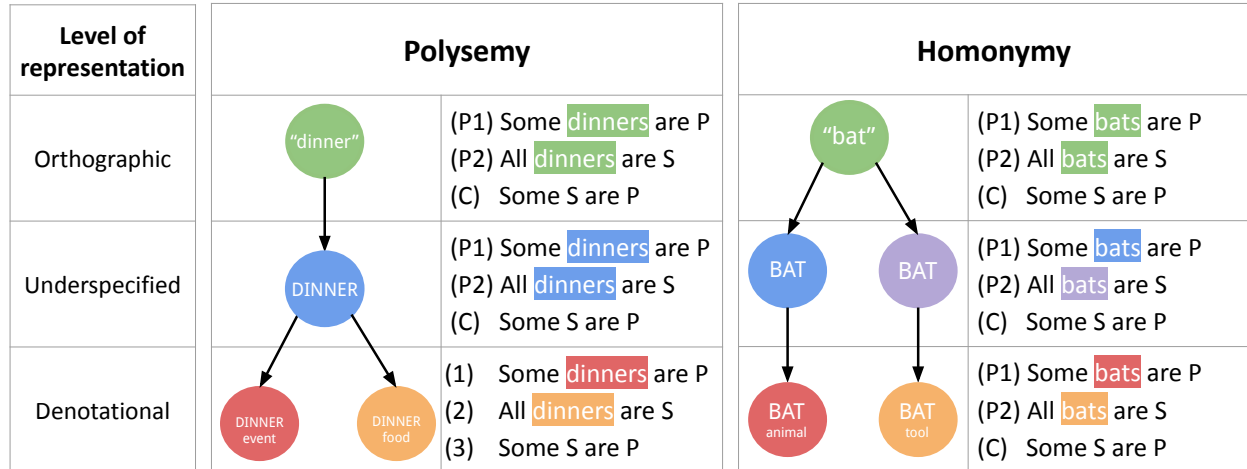


Figure 1: Illustration of how lexical ambiguity may instantiate valid deductive inference schemas at some, but not all, levels of representation. Arguments are valid when both terms are tokens of the same type (as indicated by matching colors) and invalid when they are not (mismatching colors). Both polysemous (left) and homonymous (right) arguments share the same orthographic form across premises (P1, P2) and are therefore valid if evaluated at the orthographic level. Both kinds of words equivocate at the denotational level: the premises employ different senses (e.g., dinner as food vs. event) or unrelated meanings (e.g., bat as an animal vs. tool), rendering the arguments denotationally invalid. Critically, polysemes but not homonyms share an underspecified representation which is neutral between senses, rendering the polysemy argument valid at this intermediate level. If human inference is calculated based on underspecified representations, participants should accept polysemous arguments while rejecting homonymous ones, despite their orthographic and truth-conditional equivalence.

By presenting participants with arguments that instantiate classically valid inference schemas at some, but not all, levels of representation, and by examining which conclusions they endorse, this paradigm allows us to pinpoint the kinds of representations that form the mental basis of people's capacity for deductive logical inference. Across three primary experiments (and complemented by a supplementary experiment and six norming studies reported in the Supplementary Information (SI), Sections S5-S6), we observe a robust asymmetry in participants' judgments of argument validity: arguments containing equivocal polysemes are accepted significantly more often — an overall majority — than otherwise matched arguments containing equivocal homonyms, which are largely rejected, despite being equivalent under standard truth-conditional analyses. We dub this asymmetry the *Uncommon Sense Effect* (USE).

Under the denotational assumption, USE would appear to reflect a puzzling “fallacy” of human reasoning. Polysemous and homonymous arguments alike equivocate at the level of reference, and thus neither should license valid deductive inference. This appearance of irrationality, however, rests on an overly restrictive assumption about the representations over which inference operates. Rather than reflecting a failure of our ability to apply inference rules, people's consistent endorsement of non-truth-preserving arguments arises from the systematic application of inference rules to underspecified representations that constrain, but do not fully determine, reference. Across a range of analyses — including ruling out alternative factors such as the belief bias⁴⁸, semantic priming, non-inferential heuristics, or pragmatic reanalysis — we show that underspecified representations are the default units of human deductive inference. These findings call into question a

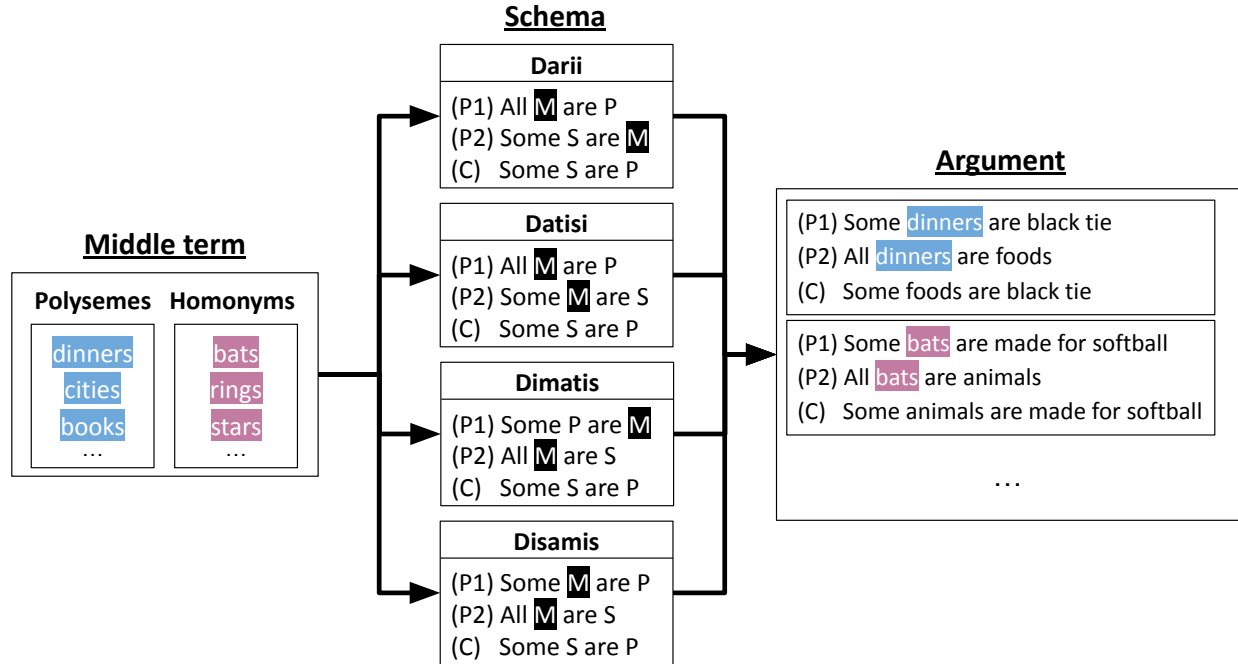


Figure 2: Diagram illustrating how critical arguments were generated. Twelve polysemes and twelve homonyms (left) were used as middle terms (M) in four Aristotelian syllogism schemas (middle; see Methods). Each middle term was used in a different sense across each argument’s two premises (right; e.g., “dinner” is used in the event sense in (P1) and the food sense in (P2)). In Experiment 1, ambiguous middle terms and syllogism schemas were fully crossed: each middle term was used in all four schemas. In Experiment 2, each ambiguous middle term was assigned to one of the four syllogism schemas with the baseline plausibility of conclusions equated across argument conditions.

foundational assumption shared across the cognitive science of meaning, reasoning, and concepts. This opens new empirical routes for investigating the representational foundations of human reasoning and new perspectives for understanding communication breakdowns, designing intelligent systems, and modeling the computational architecture of thought.

Results

Experiment 1: Establishing the Uncommon Sense Effect

In Experiment 1, participants ($N = 99$) evaluated the validity of syllogistic arguments that either equivocated over polysemous or homonymous middle terms, alongside valid and invalid filler arguments using unambiguous terms. Each middle term appeared in four separate syllogistic schemas across the experiment (Figure 2) for a total of 56 arguments per participant. Participants readily distinguished structurally valid (92% accepted) from invalid filler (40% accepted) arguments ($\chi^2(4) = 72.05, p < .001$; baseline rates which align with prior work on inference^{17,48–50}; SI Section S1). Having established that participants were sensitive to deductive validity in unambiguous filler trials, we turn to the critical comparison between polysemous and homonymous arguments. Counter to the predictions of the denotational assumption, participants judged argu-

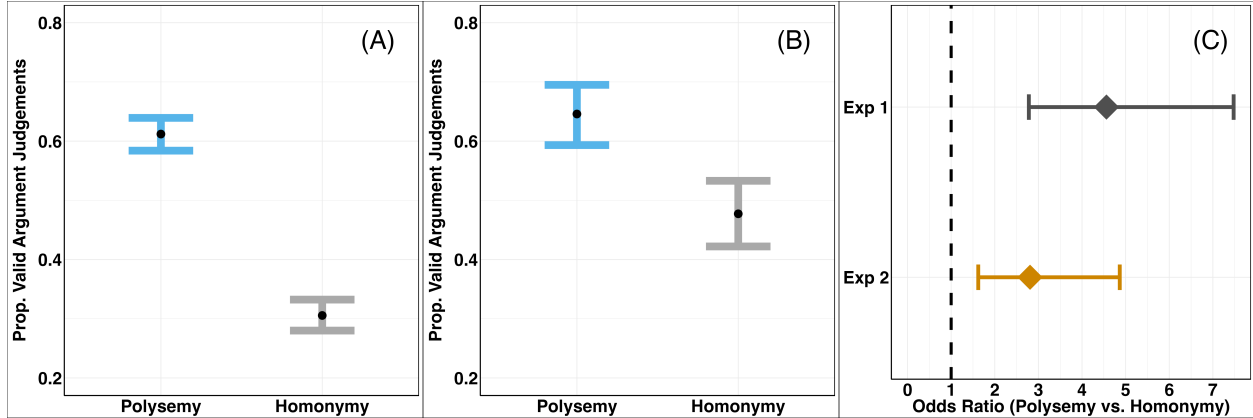


Figure 3: Left (A): proportion of arguments judged valid in the polysemous and homonymous conditions in Experiment 1. Points show mean trial-level acceptance rates; error bars indicate 95% confidence intervals. Middle (B): mean trial-level acceptance rates of polysemous and homonymous arguments (with 95% confidence intervals) in Experiment 2, where conclusion plausibility was matched across conditions. Right (C): Odds ratios (with 95% confidence intervals) for the Uncommon Sense Effect estimated from mixed-effects logistic regressions in Experiments 1 and 2.

ments to be valid on a majority of polysemy trials (61%). They did so substantially more than they accepted structurally identical arguments with equivocating homonyms (31%); a robust asymmetry visualized in Figure 3A ($\chi^2(1) = 27.3, p < .001$; Supplementary Table S1), and stable across syllogistic schemas (no interaction: $\chi^2(3) = 2.65, p = 0.449$).

This empirical pattern is inconsistent with several accounts of the representations underlying deductive reasoning. If participants' inferences operated over orthographic forms, then arguments containing polysemous and homonymous middle terms — both of which share a word form across premises (Figure 1) — should be accepted at similarly high rates. Instead, participants accepted homonymous arguments at a lower rate even than structurally invalid filler arguments (31% vs. 40% of trials). Conversely, if inference operated over denotations, then both argument types should be broadly and comparably rejected, as polysemous and homonymous arguments alike equivocate at the level of reference. Rather, participants exhibited a large and reliable asymmetry. In the best-fitting mixed-effects regression — controlling for covariates (see Methods) — polysemous arguments were accepted at over four and a half times the rate of homonymous arguments ($OR = 4.56$, 95% CI [2.78, 7.48]). We refer to this newly uncovered asymmetry as the Uncommon Sense Effect (USE).

USE generalizes across multiple levels. Although individuals varied widely in overall acceptance rates ($r = .59$ correlation between polysemy and homonymy acceptance), we nevertheless observed a reliable participant-level advantage for polysemous arguments. To assess whether USE reflects a global property or a subset of people applying a divergent reasoning strategy, we computed a participant-level USE score (see Methods). The vast majority of participants (86%) exhibited a positive USE score ($M = 0.31, SD = 0.24$, Figure 4A). Gaussian mixture modeling strongly favored a single-component solution ($\Delta BIC = 8.6$), indicating USE reflects a population-wide phenomenon rather than distinct subgroups. USE also generalized across lexical items: all twelve polysemous middle terms showed an “argument boost” ($M = 0.249, SD = 0.118$) — being judged valid at rates exceeding their conclusions presented in isolation (Figure 4B).

What else might drive USE? One possibility is pragmatic reinterpretation of ambiguous arguments. The additional context provided by the full argument may bring to mind non-default

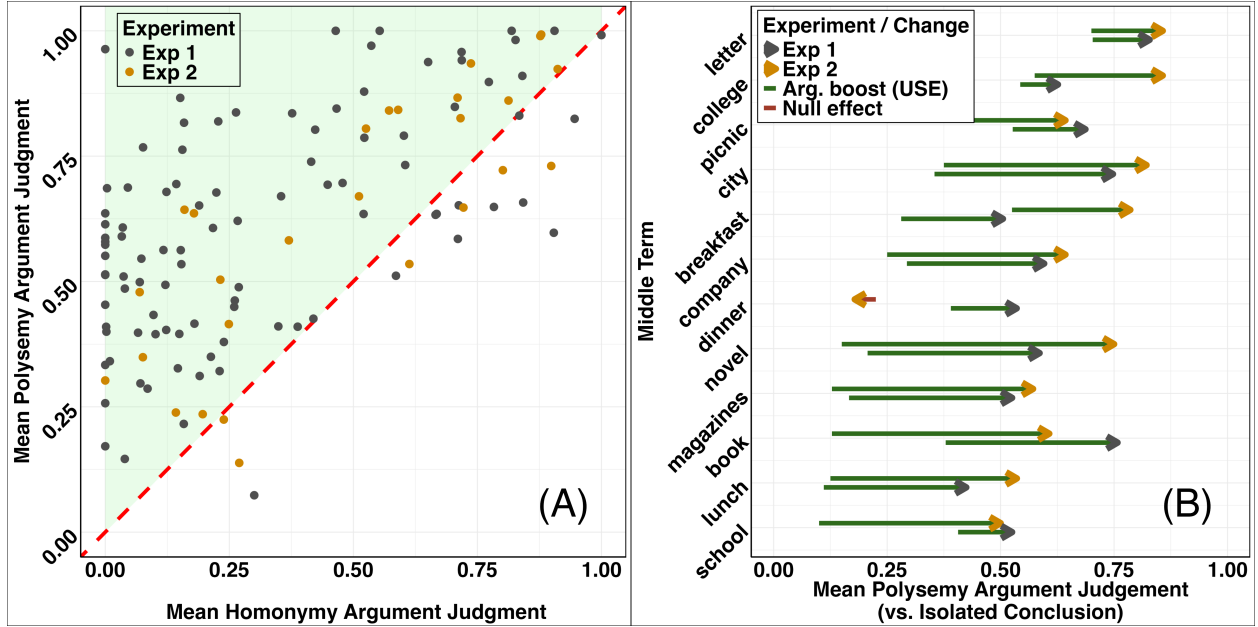


Figure 4: Left (A): Participant-level acceptance rates for polysemous vs. homonymous arguments in Experiments 1 (gray) and 2 (gold). Each point corresponds to a single participant, plotted according to their mean acceptance rate for homonymous (x-axis) and polysemous arguments (y-axis). The dashed red line indicates equality. Points above the diagonal (green shaded region) indicate a participant-level Uncommon Sense Effect. Although individuals vary widely in their overall tendency to accept arguments, the relative advantage for polysemous arguments holds broadly across participants in both experiments. Right (B): Item-level “argument boost” for each polysemous middle term, defined as the difference between the mean acceptance rate of arguments containing that term and the mean truth judgment of the corresponding conclusion presented in isolation. Arrows indicate the direction and magnitude of change, with green lines marking an argument boost (consistent with USE) and red arrows indicating a null or reverse effect. The polysemous argument boost was observed across lexical items in both Experiment 1 (gray arrows) and Experiment 2 (gold arrows).

meanings in a way which positively affects participants’ judgments. Multiple analyses rule this out: premise plausibility ratings did not differ between conditions nor improve arguments (SI, Section S7) and response times showed no evidence of effortful coercion (SI, Section S8). Furthermore, if participants gradually learned to apply pragmatic reinterpretation strategies specifically to polysemous arguments over the course of the experiment, the gap between polysemous and homonymous acceptance should widen across trials. There was no evidence for this: permissive responding did not increase over the course of the experiment and trial number does not reliably interact with USE ($\chi^2(1) = 0.01, p = .919$) nor even reliably correlate with participant behavior ($\chi^2(1) = 2.90, p = .088$, see SI, Sections S2, S8). Alternative coercion accounts that appeal to premise ordering effects (e.g., coercing the second premise to match the sense established in the first premise) make further predictions which are not borne out (SI, Section S8).

USE also cannot reflect a breakdown of structural reasoning, i.e., behaving in a manner that is insensitive to argument form. This is confirmed by the relationship between performance on filler trials and USE magnitude. Participants who best discriminated valid from invalid fillers showed twice the USE magnitude of poor performers (SI, Section S9). The fact that USE covaries

positively with sensitivity to formal structure indicates that it reflects the systematic operation of the reasoning system, rather than a breakdown in performance or adoption of a permissive response strategy. Taken together, the empirical data do not permit the attribution of USE to a subset of stimuli or participants, nor to task adaptation, pragmatic reinterpretation, generalized permissiveness, or a breakdown in the application of structural inference rules. Instead, USE is consistent with participants systematically applying valid inference schemas to underspecified representations.

Finally, although we found no interaction between judgment of validity and syllogism schema for ambiguous arguments, participants' responses were nevertheless sensitive to the plausibility of individual conclusions. Conclusion plausibility (mean conclusion truth judgment from an independent norming task, see SI S5) was a strong predictor of validity judgments, second only to Polysemy in magnitude ($OR = 2.03$, 95% CI $[1.73, 2.38]$). This is consistent with previous research on inference in which participants exhibit a "belief bias," accepting arguments more often when their conclusions are more believable^{17;48;49}. As a consequence of stimulus construction, conclusions in polysemous arguments were, on average (normed conclusion judgments: $M = 0.363$, $SD = 0.255$), more plausible than those in homonymous arguments ($M = 0.166$, $SD = 0.166$). To address this potential confound, in Experiment 2 we constructed a new set of arguments in which conclusion plausibility was matched across polysemous and homonymous stimuli, based on independent norming. This design experimentally controls for the belief bias. It allows us to assess whether—and to what extent—the Uncommon Sense Effect observed in Experiment 1 persists when differences in conclusion plausibility are eliminated, thereby testing whether USE reflects underspecified representations rather than differences in propositional plausibility.

Experiment 2: Controlling for belief bias

Experiment 2 tested whether USE persists when conclusion plausibility is experimentally controlled. Participants ($N = 28$) evaluated the same argument types as Experiment 1, but with conclusions matched for plausibility across polysemous and homonymous conditions (both $M = 0.31$; SI Section S5.3). Because Experiment 1 revealed no Polysemy $\times$ Schema interaction, each middle term appeared in a single schema rather than being fully crossed with all four, facilitating plausibility matching while preserving inferential structure.

As in Experiment 1, participants readily distinguished structurally valid (91% accepted) from invalid filler (46% accepted) arguments involving unambiguous terms ($\chi^2(1) = 30.08$, $p < .001$); confirming participants remained sensitive to deductive validity. Turning to the core theoretical comparison, participants judged polysemous arguments to be valid more often than structurally matched homonymous arguments (65% vs. 48%; see Figure 3B); a statistically significant difference ($\chi^2(1) = 10.86$, $p < .001$; Table S9) that cannot be attributed to belief bias since the plausibility of conclusions was equated across critical conditions. In fact, Polysemy remained the single strongest predictor of argument validity judgments in the best-fitting mixed-model ($OR = 2.81[1.62, 4.87]$). Thus, USE persists even when differences in conclusion plausibility are eliminated.

Additional analyses replicated the generality of USE observed in Experiment 2. The effect was robust across participants, with 82% exhibiting a positive participant-level USE score. Gaussian mixture modeling again favored a single-component solution ($\Delta BIC = 3.5$), providing no evidence for distinct subpopulations. USE also generalized across lexical items: in Experiment 2, eleven out of twelve polysemous middle terms exhibited argument boost relative to their arguments' conclusions presented in isolation ($M = 0.337$, $SD = 0.173$; see Figure 4).

In addition to controlling for conclusion plausibility, we further tested whether USE depends on reflective, or resource-intensive reasoning. In an additional experiment (SI, Section S6), participants

($N = 118$) completed an argument validity task using the same stimuli from Experiment 2 but placed under various kinds of cognitive load (induced via time pressure or concurrent digit memorization, each at varying difficulty levels). Despite substantially reduced opportunity for deliberation, USE was observed: polysemous arguments continued to be accepted at significantly higher rates than homonymous arguments ($\chi^2(1) = 5.86, p = 0.015$). The persistence of USE under load further rules out accounts on which the effect arises from reinterpretation or strategic reasoning. Instead, USE seems to reflect a default, automatic inferential process which operates over underspecified representations.

One remaining possibility concerns semantic relatedness — a property closely associated with polysemy. Polysemous senses are characteristically related whereas homonymous meanings are arbitrary. Consequently, arguments containing polysemous middle terms may involve greater overlap in semantic content across premises relative to homonymous arguments. It is possible that USE stems from this semantic association, rather than a structural property of words in the mind. Importantly, the experiments thus far rule out explanations based solely on shared orthographic strings: polysemous and homonymous arguments both repeat the same word form across premises, yet only the former are reliably endorsed. What remains unresolved is whether semantic relatedness alone — independent of lexical identity — can support deductive inference. Experiment 3 directly tests this possibility.

Experiment 3: Ruling out semantic relatedness

In Experiment 3, we tested whether USE could be explained by relatedness between meanings, rather than by inference over underspecified representations. We constructed arguments in which the repeated middle term was replaced by pairs of lexical items that are highly semantically related but — due to their distinct word forms — do not share a single underspecified representation. Participants ($N = 31$) evaluated syllogistic arguments in four conditions. In addition to structurally valid and invalid filler items, the critical conditions were *Related* and *Unrelated* word pair arguments. Arguments in the Related condition contained semantically related but lexically distinct word pairs across premises (e.g., “socks” and “shoes”) ⁵¹, rather than equivocating over a single polysemous or homonymous term (Figure 5). Arguments in the Unrelated condition were structurally and conceptually identical to the Related ones except that the pair of would-be middle terms was devoid of clear semantic connection (e.g., “lions” and “religions”). If semantic relatedness is sufficient to drive USE, Related arguments alone should pattern like polysemous ones and be broadly accepted. If, instead, USE depends on inference over a shared, underspecified representation, then both Related and Unrelated arguments should pattern like homonymous ones and be rejected. This design isolates semantic relatedness from lexical identity, providing a critical test of the underlying cause of USE.

As before, participants distinguished structurally valid (87% accepted) from invalid filler (34% accepted) arguments involving unambiguous repeated middle terms ($\chi^2(1) = 47.29, p < .001$). Invalid filler trials also provide a baseline for comparison with Related and Unrelated arguments. In contrast with Experiments 1 and 2 where participants systematically endorse non-truth-preserving arguments containing equivocal polysemes, critical arguments in Experiment 3 were consistently rejected. Both the semantically-Related items (25%) and the semantically-Unrelated items (10%) were accepted less frequently than the structurally invalid Fillers (34%)

There was a descriptive advantage for Related items ($\chi^2(1) = 6.93, p = 0.008$), however, this tendency appears to be driven by the belief bias: Related arguments in our stimuli had higher average conclusion plausibility ($M = 0.279, SD = 0.226$) than Unrelated arguments ($M = 0.139, SD = 0.151$), a gap of 0.14 which almost exactly maps onto the validity judgments. This is perhaps

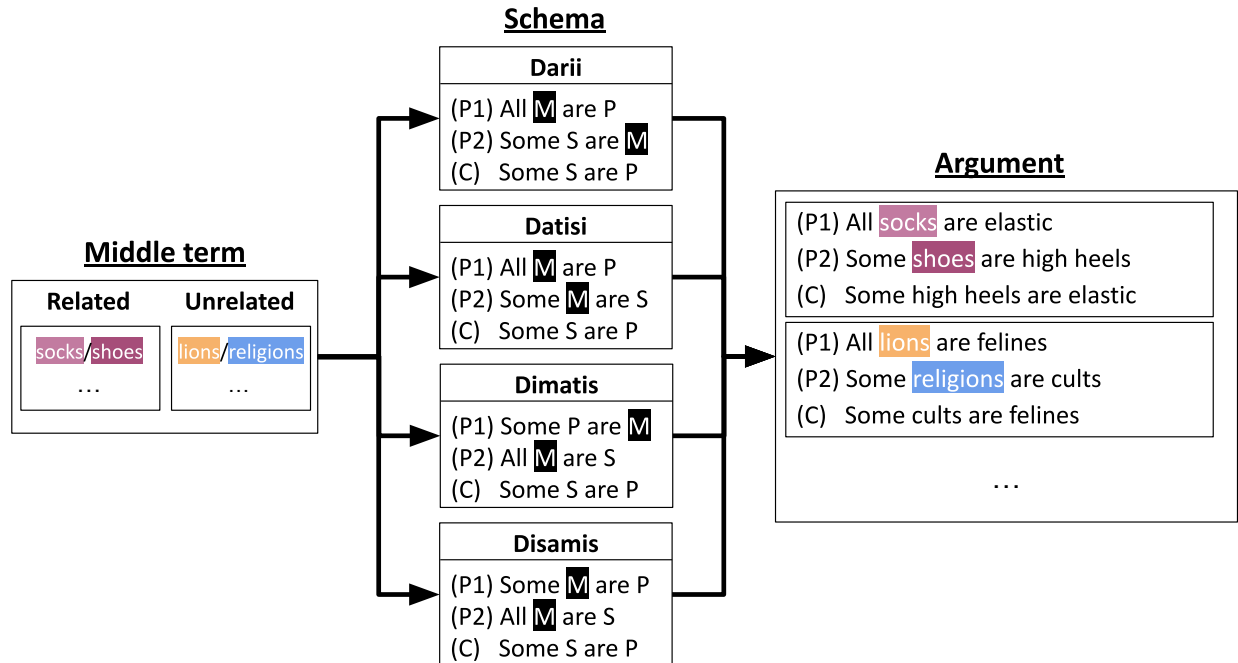


Figure 5: Diagram illustrating how Experiment 3 arguments were generated. We selected pairs of either semantically related words (12 pairs) or unrelated words (12 pairs), based on independent measures of relatedness^{51,52} (see Methods). Each word pair was then placed into the middle term position (M) of one of four Aristotelian argument schemas (see Methods). Based on this constraint, plausible premises were written, which in conjunction with the argument schema, dictated the conclusions. If the USE is driven by the difference in semantic relatedness between polyseme and homonym meanings, we should see a similar difference between acceptance of arguments with semantically related versus unrelated word pairs.

most striking when viewed at the individual argument level: in marked contrast with Polysemy arguments in Experiments 1 and 2 — which were accepted at nearly double the rate of their conclusions in isolation — the Related and Unrelated items in Experiment 3 were endorsed as valid *less frequently* than their isolated conclusions; a change in acceptance of -0.034 and -0.036 , respectively. Contrary to an account that attributes USE to semantic relatedness alone, when adjusted for conclusion plausibility, a simple linear regression revealed no difference in item-level argument boost between Related and Unrelated conditions ($t(22) = 0.025, p = .980$), with both slightly negative — Figure 6. Thus, even when semantic association is strong enough to produce robust priming effects (e.g., in item naming or lexical decision tasks⁵¹), it does not support deductive inference in the absence of a shared underspecified representation.

Discussion

We have identified a systematic asymmetry in human deductive reasoning: people reliably accept arguments that equivocate over polysemous terms while rejecting structurally identical arguments with homonymous terms. This Uncommon Sense Effect (USE) reveals that logical inference operates over underspecified representations — mental structures that constrain meaning but do not fully determine truth-conditional thought. This finding challenges a foundational assumption

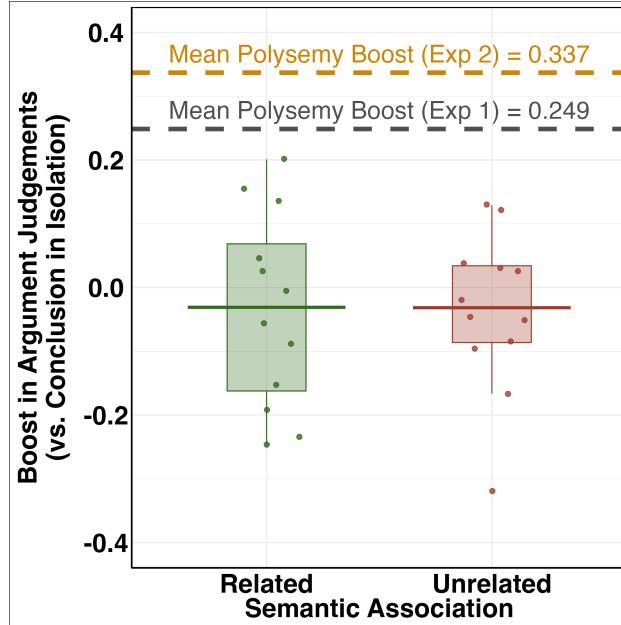


Figure 6: Distribution of item-level “argument boost” in Experiment 3, defined as the difference between the mean acceptance rate of an argument and the mean truth judgment of its conclusion presented in isolation. Each point corresponds to a single argument constructed with either a semantically related (green) or semantically unrelated (red) word replacing the equivocal middle term. Each point represents a single argument item; boxes show inter-quartile ranges for each condition. Dashed horizontal lines indicate the mean polysemy boost observed in Experiments 1 (grey) and 2 (gold) for comparison.

linking meaning, truth, and reasoning across cognitive science.

Independent of theoretical commitments, USE represents a novel and robust psychological phenomenon which demands explanation. Any adequate theory of human reasoning must account for why structurally identical arguments containing different types of lexical ambiguity produce such divergent patterns of endorsement. The challenge is not whether to explain USE, but how. In this respect USE—and the apparent tension between non-ideal performance and systematic underlying competence—is akin to well-known reasoning biases, such as the conjunction fallacy^{53;54} or the gambler’s fallacy⁵⁵ (cf.⁵⁶). There, too, as in learning and processing more broadly^{57–61} systematic but ostensibly irrational behavior reflects the proper functioning of previously unknown representational systems.

Nevertheless, our empirical results strongly constrain viable alternatives. USE cannot be attributed to the plausibility of individual premises (no differences across conditions) or conclusions (controlled in Experiment 2), semantic relatedness (Experiment 3), pragmatic reinterpretation or coercion of different senses (SI, Sections S6 and S8). Moreover, participants who performed best at discriminating structural validity showed the largest USE (SI, Section S9), indicating the effect reflects systematic application of inference rules rather than a breakdown in reasoning. These patterns are inconsistent with reasoning over either orthographic forms (homonyms would be accepted) or denotations (polysemes would be rejected). While we are open to alternative explanations of the phenomenon, the data strongly suggest that underspecified representations provide the most parsimonious account. Polysemous words possess a shared representation that is neutral between senses and licenses inference, while homonyms do not.

While our results indicate that underspecified representations are the basis of deductive reasoning, the data are consistent with multiple views of their precise nature. A representation may be underspecified because it encodes less information than is required to represent truth or falsity^{38;62}—perhaps as minimal as a pointer to a memory location—or because it encodes more information than required. On the latter, a subset of word information must be selected to represent truth or falsity⁶³. These two views correspond to what have been termed “thin” and “rich” respectively³⁶, both of which are compatible with the present findings. The “thin” view in particular finds a natural analog in underspecification applied to other domains such as phonological representation and speech processing^{64;65}.

Our results are not, however, consistent with every existing representational account of lexical ambiguity. Distributed semantic models (including contemporary language models) characterize meanings as distributions of co-occurrence or feature activation^{66–69}, with polysemes and homonyms as differing only in degree: a polyseme’s senses contain many overlapping semantic features, whereas a homonym’s meanings contain few, if any. Yet if USE were driven solely by graded similarity, semantically related word pairs in Experiment 3 should have patterned with polysemes. Instead, they patterned with homonyms, despite robust semantic association. This suggests the polysemy-homonymy distinction cannot be reduced to merely the density of overlapping features. A more direct test may come from looking at different classes of polysemy. Our stimuli rely on the widely discussed “inherent” polysemes⁷⁰ which support copredication across its senses^{63;71}. While evidence of underspecification is strongest in “inherent” polysemy^{45;46;72}, it is an open question whether USE extends to other^{71;73} instances of polysemy as well.

A related question concerns how learners acquire the structural distinction between polysemy and homonymy. Part of learning a word must involve implicitly learning whether its form maps to a single underspecified representation or to multiple unrelated meanings. Children as young as four years old extend novel meanings for regular-polysemous relations but not homonyms in a learning context^{47;74}. Whether this learning is driven primarily by semantic overlap or by rule-like generalization over lexical types, as seen in other aspects of language acquisition^{75;76}, remains an open question.

A further question concerns the scope of processes in which underspecified representations are used. Thus far, we have focused on deductive syllogisms; it remains to be seen whether underspecified representations play a comparable role in other forms of inference. One natural extension is to non-syllogistic deduction, such as propositional inference patterns governed by rules like *modus ponens* or disjunctive syllogism. Beyond deduction, underspecified representations may also participate in inductive reasoning, where conclusions extend probabilistically beyond the evidence. Exploring these additional inferential domains would clarify whether underspecified representations constitute a domain-general basis for human reasoning or are restricted to particular inference types. If USE-like patterns emerge across diverse reasoning tasks, this would suggest that underspecification is not merely a feature of how we process ambiguous language, but a fundamental component of human thought. Independent of reasoning, underspecified representations may also play a role in communication. Interlocutors can plausibly sustain coherent exchanges without converging on identical contents. Perhaps they do so by relying on underspecified representations, and breakdowns can arise if further, specific meanings diverge.

Finally, while we have argued that underspecified representations are the basis of logical inference, we do not deny that humans can use denotations to represent thoughts that can be true or false or that inferences can show sensitivity to truth. Clearly people are capable of recognizing that polysemes have multiple distinct senses. The critical question is which representational level serves as the default for deductive reasoning—and under what conditions might people resolve to a non-default representation. Several lines of evidence suggest underspecification

as the default: USE shows no reduction as a function of task exposure, persists under time pressure, and correlates positively with structural reasoning ability. It is likely that accessing fully specified denotations, in fact, requires additional resources or relies on task demands beyond what the present paradigm elicits on most trials. Nevertheless, our establishment of USE and the underspecified representations that support it are important steps in understanding how and why people may reason using different potential representations.

Given these findings, we return to the denotational assumption: the widespread view that inference operates over the same representations that determine truth. If underspecified representations — not denotations — are the default basis for reasoning, then the denotational assumption requires substantial revision. This carries wide implications across psychology, linguistics, and philosophy. Consider first the theory of concepts. Psychology typically conceives of concepts as playing two roles: they are the basic units over which people infer in domain-general reasoning, and they are the fundamental bearers of denotations, allowing thoughts to bear truth or falsity^{24;77;78}. Our results suggest that the representations used in inference are not, by default, bearers of truth. This raises a dilemma: either concepts are the basic units of inference but do not carry denotations, or they are the fundamental bearers of denotations but are not used in deductive reasoning. No single mental structure appears to play both roles. Perhaps progress can be made by dividing and conquering.

Formal linguistics similarly operates under the denotational assumption: entailment relations hold in virtue of the preservation of truth^{25;27;30}; cf.⁷⁹. On this view, entailment relations are defined over denotational representations. Yet our findings reveal that people systematically judge entailment relations to hold between non-denotational, underspecified representations. One could maintain the denotational assumption despite this tension by treating it as a productive idealization. Idealization is ubiquitous in scientific practice from frictionless planes in physics⁸⁰ to ideally rational agents in economics⁸¹ — assumptions justified to focus on relevant phenomena under investigation⁸² or to make difficult problems tractable^{80;83}. This undoubtedly holds in paradigmatic cases where underspecified and denotational representations hold in one-to-one correspondence, or where no equivocation exposes their structural distinctness.

However, treating the denotational assumption as mere idealization comes at a cost. Full understanding of any complex system must ultimately take the form of causal components and mechanisms out of which it is constructed, not just input-output trends^{84;85}. When studying human cognition, this means identifying the actual mental representations and processes that generate behavior. The present findings demonstrate this concretely: once we look beyond truth, previously obscured structure in human reasoning becomes both theoretically visible and empirically tractable.

Methods

Participants

Participants were recruited online via Prolific⁸⁶ and compensated at a rate of approximately \$15 per hour. All participants were native English speakers residing in the United States who had completed at least 15 prior submissions on the platform. Each participant completed only a single experiment or norming study. All studies were preregistered: Experiment 1 on April 18 2025, Experiment 2 on December 10th 2024, and Experiment 3 on May 27th 2025. As preregistered, participants were excluded if their mean response time across all trials was more than two standard deviations above or below the sample mean, or they failed to respond on at least 10% of trials.

For Experiments 1 and 3, we additionally excluded participants who accepted more than 70% of structurally invalid filler arguments; this criterion was preregistered at 50% but adjusted to 70% prior to data analysis to be more inclusive while still identifying inattentive responding though this is not qualitatively affect the results.

A total of 158 participants contributed data to the three primary experiments — 99 to Experiment 1, 28 to Experiment 2, and 31 to Experiment 3 — after applying preregistered exclusion criteria (exclusion rate of 15.5% from the total recruited 187 participants). An additional 412 participants contributed data to the norming studies and the cognitive-load experiment reported in the Supplementary Materials (exclusion rate of 1.7% from the total recruited 419 participants). All experiments were approved by the City University of New York University Integrated Institutional Review Board (Protocol #2024-0470-GC), and all participants provided informed consent prior to participation.

Design

All three experiments tasked participants with evaluating the validity of deductive syllogistic arguments. Each experiment employed a mixed design with Trial type (e.g., Polysemy, Homonymy, Filler, etc.) as a within-subjects factor and stimulus list/order as a between-subjects factor. Trial types were comprised of “critical” and “filler” arguments. Items in the critical condition were structurally valid but equivocated over the meanings of either lexically ambiguous middle terms (Experiments 1 and 2) or semantically controlled word pairs (Experiment 3). Items in the filler condition contained lexically unambiguous middle terms but instead manipulated structural validity; thus being divided into Valid Fillers and Invalid Fillers. All stimulus lists consisted of 56 arguments evenly divided among four Aristotelian syllogistic schemas (Dimatis, Disamis, Datisi, and Darii as in Figure 2; 14 arguments per schema).

In Experiment 1, middle terms and syllogistic schemas were fully crossed such that each middle term appeared in each of Dimatis, Disamis, Datisi, and Darii exactly once across four unique lists. Each list was then put in two randomized orders to create eight between-subjects conditions with each participant thus judging 12 polysemous arguments, 12 homonymous arguments, 16 structurally valid fillers, and 16 structurally invalid fillers.

Experiments 2 and 3 each used a single stimulus list assigned to participants in one of two randomized orders. Based on results from Experiment 1 showing no interaction between the critical condition (Polysemy vs. Homonymy of the middle term) and syllogism schema, in Experiments 2 and 3 each middle term was randomly assigned to (rather than crossed with) one of the four schemas to generate an argument. Stimuli in Experiment 2 consisted of 12 polysemous and 12 homonymous arguments along with the same 32 filler (structurally valid/invalid) arguments as Experiment 1. Given the effect of isolated conclusion plausibility on validity judgments (i.e., belief bias), we matched the mean conclusion plausibility of the critical trial types (Polysemy and Homonymy) in Experiment 2 based on independent norming data (SI Section S5.3). Experiment 3 used 12 semantically-related word pair arguments and 12 semantically-unrelated word pair arguments along with the same 32 filler arguments. Apart from replacing premises containing lexically ambiguous middle terms with semantically-related or -unrelated word pairs, the design of Experiment 3 was the same as Experiment 2.

Materials

All stimuli were Aristotelian syllogisms consisting of two premises sharing a middle term (M) along with a conclusion. Although there are 15 unconditionally logically valid syllogistic schemas,

we restricted our materials to four based on real-time processing considerations. First, all arguments contained only affirmative statements (e.g., “All M are P”), as these are generally processed faster and more reliably than negated statements (e.g., “No M are P”) ^{8;87}. Second, we only employed arguments with existentially quantified conclusions (e.g., “Some M are P”) since universal conclusions (e.g., “All M are P”) are rendered invalid by a single counter-instance, and are thus less plausible in general. Applying these constraints yields the four syllogistic schemas used throughout the study, referred to by medieval logicians as Dimatis, Disamis, Datisi, and Darii ^{88;89}.

Middle Terms

The twelve polysemous middle terms used in Experiments 1 and 2 were regular polysemous nouns ⁷⁰ — words showing a systematic relation in meaning that recurs across the lexicon — drawn from Murphy (2021) ⁷². These terms span three widely discussed classes of polysemy, distinguished by the relation between their senses: book-type polysemy (“book,” “magazine,” “letter,” “novel”), ambiguous between physical objects and information content (e.g., “paperback novel” vs. “romance novel”); lunch-type polysemy (“lunch,” “dinner,” “picnic,” “breakfast”), ambiguous between physical foods and events (e.g., “overcooked breakfast” vs. “brief breakfast”); and city-type polysemy (“city,” “company,” “school,” “college”), ambiguous between institutions and physical locations (“dilapidated school” vs. “prestigious school”). Including multiple relation types ensured that any observed effects reflect properties of the general representation of polysemous words, and are not due to idiosyncrasies of a particular relation (e.g., book-type only) or individual lexical item (e.g., just “book”).

Ten of the twelve homonymous middle terms (“board,” “suit,” “cabinet,” “bat,” “ball,” “ruler,” “ring,” “match,” “star,” “change”) were adapted from Frazier & Rayner (1990) ⁴³, with two additional homonyms (“wave” and “rock”) added to increase lexical diversity and to equate the number of polysemous and homonymous terms across conditions.

Middle terms of filler arguments were unambiguous nouns selected to match the critical items in average length and frequency. We identified candidate filler terms from the SUBTLEX-US database ⁹⁰ by filtering for morphologically regular plural nouns (i.e., ending in “-s”) that fell within the inter-quartile range of the polysemous and homonymous middle terms for both word frequency ($\log_{10}$ frequency: 1.62-15.20) and word length (between six and nine letters). From the resulting candidate list we manually selected 32 terms, excluding proper nouns and words which may ambiguously be interpreted as verbs (e.g., “regrets”)

In Experiment 3, the polysemous and homonymous middle terms were replaced with semantically-related and semantically-unrelated word pairs. To align with the polysemy and homonymy conditions as closely as possible, these pairs consisted of category-denoting common nouns. The members of each pair referred to disjoint sets — no word pair stood in hierarchical relation (e.g., “food” and “hamburger”) — mirroring the disjoint extensions of polysemous senses and homonymous meanings in Experiments 1 and 2.

These pairs were selected from the Semantic Priming Project (SPP) ⁵¹, a large-scale database of word pair priming responses, and free-association norms ⁵² based on two criteria. First, semantically-related items must comprise a cue and its most commonly produced first-associate word ⁵². Second, related word pairs demonstrated a priming magnitude in the top 500 largest across both a primed lexical decision task and a primed speech-onset naming task. These criteria ensured that related-pairs were both intuitively associated, while also exhibiting strong semantic priming across multiple behavioral tasks. The twelve semantically-unrelated pairs were randomly selected from the SPP control condition, thus exhibiting no measurable priming and lacking any intuitive semantic relationship.

Arguments

Critical arguments were generated by slotting in a middle term (M) from the above sets into a given syllogism schema. Premises were written using end terms¹⁷ P and S such that each proposition biases the interpretation of the ambiguous middle term towards a different sense or meaning. Premises were written such that the first premise biased the middle term towards its subordinate sense, while the second premise biased towards the dominant sense; with dominance status determined based on findings from a previous sentence completion task⁷². For instance, for an initial premise like “Some magazines are coffee-stained” readers should adopt the subordinate physical-object sense, thus the second premise “All magazines are subscriptions” biases towards the more common information-content sense. Given a middle term and the end terms from the two premises the form of the conclusion sentence is defined automatically by the syllogism schema (e.g., “Some subscriptions are coffee-stained”)

Structurally valid filler arguments were constructed by faithfully applying each syllogism schema, whereas structurally invalid filler arguments swapped the order of terms in the universal premise (e.g., converting “All P are M” into “All M are P”). This procedure yields valid and invalid arguments in all four syllogistic schemas.

The stimuli used in Experiment 1 consisted of 224 unique arguments generated by slotting each of the twelve polysemous, twelve homonymous, and thirty-two unambiguous middle terms into each of the four syllogistic schemas. These were divided into between-subjects lists as described above (see full argument set in SI; Table S25)

Since the purpose of Experiment 2 was to control for any effects of the belief bias (differences in plausibility of isolated conclusions across conditions), we first wrote multiple versions of arguments for each middle term (using different end terms S and P), and based on the results of a norming study (SI Section S5.3) chose one argument per middle term such that the polysemy and homonymy items were precisely matched for mean truth judgments on their conclusions (see SI; Table S26 for full set). For Experiment 2, a typographical error discovered after data collection required exclusion of one homonymous argument, resulting in eleven homonymous arguments and twelve polysemous arguments for this experiment. Construction of arguments for Experiment 3 followed the same general procedure, except using semantically-controlled word pairs as above (SI; Table S27).

Procedure

Experiments were coded using PCIBex⁹¹ and participants completed the study in their web browser. At the beginning of the study, participants were told to assume that the premises of each argument were true and then to judge whether the conclusion follows from the premises. This instruction was intended to ensure that participants focused on arguments’ validity, not the likelihood of truth of the premises or conclusions if considered in isolation — although we assessed and controlled for the effects of premise and conclusion plausibility via independently gathered norming data (SI Sections S5.1-S5.6). Participants were then provided with a brief explanation of validity and some examples of valid and invalid arguments. Sentences in each argument were presented in sequence, so that participants only saw one statement on the screen at a time, pressing a key to advance. This self-paced sequential mode of presentation was chosen to ensure that participants could not overtly reanalyze premises in light of subsequent context. After all three statements within an argument had been displayed, participants were tasked with selecting “yes” or “no” in response to the question “Does the conclusion follow from the premises?”

Analysis

All analyses were conducted in R (v4.3.1). We fit mixed-effects logistic regression models to trial-level data using the lme4 package (v1.1.35.1), with Nelder-Mead and BOBYQA optimizers. For each experiment, we fit separate models for critical trials comparing polysemous and homonymous arguments (Experiments 1 and 2) or semantically related and unrelated word pairs (Experiment 3), as well as filler trials comparing structurally valid and invalid arguments. The dependent variable in each case was “yes” responses—whether participants accepted each argument as valid. The primary independent variable of interest was trial type; this was treatment-coded with “Homonymy” as the reference level in Experiments 1 and 2, “Unrelated” word pairs as the reference level in Experiment 3, and “Invalid Filler” as the reference level in all filler trial analyses.

All models included the following control variables as fixed effects: response time (RT), trial number, and independently normed premise plausibility ratings for both premises. For critical trial analyses, we additionally included independently normed conclusion plausibility ratings to control for belief bias effects. All continuous predictors were centered and scaled ($M = 0$, $SD = 1$). For Experiment 1, syllogism schema (Darii, Datisi, Dimatis, Disamis) was sum-coded and included as a fixed effect to test for both main effects and interactions with trial type. Based on the design of Experiments 2 and 3 along with the lack of interaction between schema and trial type ($\chi^2(3) = 2.65$, $p = 0.449$), schema was not included as a predictor in Experiments 2 and 3.

Random effect structures included by-subject random intercepts and by-item (middle term) random intercepts in all models. We additionally included by-subject random slopes for trial type when this significantly improved model fit: for filler trials for all three experiments and for critical trials for Experiment 1 only. We tested the significance of factors using likelihood-ratio tests on the χ^2 values from nested model comparisons with identical random effect structures. We report regression coefficients with 95% confidence intervals, along with odds ratios for all fixed effects.

To assess individual differences in response patterns, we computed Pearson correlations between participants’ mean acceptance rates for polysemous and homonymous arguments. We also calculated an “argument boost” score for each middle term, defined as the difference between the mean acceptance rate of arguments containing that term and the mean normed truth judgment of that term’s conclusion when presented in isolation. For Experiment 3, we additionally conducted item-level linear regression analyses to test whether argument boost differed between Related and Unrelated conditions after controlling for conclusion plausibility.

Finally we examined whether the Uncommon Sense Effect (USE) reflects a population-wide tendency or arises from a distinct subgroup applying a divergent reasoning strategy, we computed participant-level USE scores for each experiment—defined as the difference between each participant’s mean acceptance rate for polysemous and homonymous arguments (or related and unrelated). To test whether individual differences in USE scores reflected a single homogeneous population or multiple distinct subgroups (e.g., participants who rely on underspecified representations versus those who do not), we fit Gaussian finite mixture models to the distribution of participant-level USE scores using the mclust package (v6.0.0). We performed Bayesian Information Criterion (BIC)-based model selection from models with 1 to 5 components. For models with multiple components, both equal-variance and variable-variance specifications were tested, with the lowest BIC value indicating the best-fitting model.

Data availability

All trial-level data and pre-registrations supporting the findings of this study are made freely available via the Open Science Framework at <https://osf.io/mav8s/>.

Code availability

All code for pre-processing, visualization, and statistical analysis is available via GitHub at <https://github.com/scaplan/uncommon-sense-effect>.

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Supplementary Information:

“The Uncommon Sense Effect: experimental evidence for underspecification as the mental basis of logical inference”

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S1 Baseline Acceptance Rates for Structurally Invalid Arguments

As evidence that participants are sensitive to structural validity in our deductive inference paradigm, we note that participants reliably discriminate valid from invalid filler arguments that do not contain any equivocation; i.e., lexically unambiguous middle terms. However, one may wonder whether the present acceptance rates for invalid fillers (Experiment 1: 40%; Experiment 2: 46%; Experiment 3: 34%) are unusually high. Such rates are, in fact, well within the expected range for adult reasoners. In the most analogous previous studies, which require adults to make binary forced validity judgments on valid and invalid syllogisms, participants accepted invalid arguments approximately 40% of the time⁴⁸; this figure (in line with similar studies, e.g.⁴⁹) is the mean of believable and unbelievable invalid acceptance from a meta-study combining the baseline acceptance rates for three previous experiments⁵⁰. Furthermore, studies that present participants with premises and ask them to deduce their own conclusion reinforce the claim that items like our structurally invalid fillers are difficult to answer correctly on all trials, despite humans' universal aptitude for inferential and deductive reasoning. In one analysis aggregating results from six independent experiments¹⁷, invalid forms were only correctly answered between 12 and 37% of the time (¹⁷ table 6; the invalid fillers in the present study correspond to these authors' syllogisms AI1 (16% correct), AI1 (37% correct), IA2 (12% correct), and IA3 (28% correct)). Thus, the baseline acceptance rate for structurally invalid filler arguments exhibited by our participants is in line with the range established in the existing literature.

S2 Detailed Analyses: Experiment 1

We fit two sets of mixed-effects logistic regression models for Experiment 1: one comparing Valid and Invalid filler arguments, and one comparing Polysemous and Homonymous critical arguments. Here we report the full model specifications and selection procedures.

S2.1 Critical Trials: Polysemy vs. Homonymy

The best-fitting model (output in Table S1) for critical trials included:

- Fixed effects:
 - Trial Type (Polysemy vs. Homonymy ; treatment-coded with Homonymy as reference)
 - Syllogism Schema (Darii , Datisi , Dimatis , Disamis ; sum-coded)
 - Response time (RT ; centered and scaled)
 - Trial number (ItemNum ; centered and scaled)
 - Conclusion plausibility (stim.ConcTruth ; centered and scaled)
 - Premise 1 plausibility (stim.premiseJudgment_P1 ; centered and scaled)
 - Premise 2 plausibility (stim.premiseJudgment_P2 ; centered and scaled)
- Random effects:
 - By-subject random intercepts and random slopes for Trial Type
 - By-item (middle term) random intercepts

Full model specification as an R formula:

Response ~ TrialType + SyllogismSimple + RT + ItemNum +
 ConcTruth + PremiseJudgmentP1 + PremiseJudgmentP2 +
 (1 + TrialType | SubjectID) + (1 | TargetWord)

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-1.12 [-1.61, -0.63]	-4.51	< .001	0.33 [0.2, 0.53]
Polysemy	1.52 [1.02, 2.01]	6.02	< .001	4.56 [2.78, 7.48]
Darii	0.3 [-0.09, 0.68]	1.52	.127	1.35 [0.92, 1.98]
Datissi	0.53 [0.14, 0.91]	2.70	.007	1.7 [1.16, 2.49]
Dimatis	-0.73 [-1.13, -0.33]	-3.57	< .001	0.48 [0.32, 0.72]
RT	0.01 [-0.11, 0.14]	0.21	.833	1.01 [0.9, 1.15]
Trial Number	-0.1 [-0.21, 0.01]	-1.73	.084	0.91 [0.81, 1.01]
Conclusion Strength	0.71 [0.55, 0.87]	8.78	< .001	2.03 [1.73, 2.38]
Premise1 Strength	-0.04 [-0.36, 0.28]	-0.25	.805	0.96 [0.7, 1.32]
Premise2 Strength	-0.43 [-0.66, -0.2]	-3.62	< .001	0.65 [0.52, 0.82]

Table S1: Output of the best-fitting model for critical trials (Polysemy and Homonymy) in Experiment 1. Bracketed values are 95% confidence intervals. $N = 2,376$ trials from 99 participants.

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit over a model with only random intercepts (Table S2).

Model	npar	AIC	BIC	logLik	χ^2	p
Random Intercepts Only	15	2542.4	2629.0	-1256.2	–	–
Random Slopes	17	2498.5	2596.7	-1232.3	47.86	<.001

Table S2: Comparison of random effects structures for critical trials (Polysemy vs. Homonymy) in Experiment 1.

The random slopes model provided significantly better fit and was retained for subsequent comparisons. Next, we tested whether the interaction between Trial Type and Syllogism Schema improved model fit (Table S3).

Model	npar	AIC	BIC	logLik	χ^2	p
Main Effects Only	14	2495.2	2576.0	-1233.6	–	–
With Interaction	17	2498.5	2596.7	-1232.3	2.65	.449

Table S3: Test of Trial Type $\times$ Schema interaction for critical trials (Polysemy vs. Homonymy) in Experiment 1.

The interaction term did not significantly improve fit and was dropped from the final model. Finally, we confirmed the main effect of Trial Type by comparing the full model with Trial Type to a nested model without it (Table S4).

S2.2 Filler Trials: Structurally Valid vs. Invalid

Conclusion plausibility was excluded from filler models because it was not statistically separable from structural validity. In filler trials, structurally invalid arguments systematically featured

Model	npar	AIC	BIC	logLik	χ^2	<i>p</i>
Without Trial Type	13	2520.5	2595.5	-1247.2	–	–
With Trial Type	14	2495.2	2576.0	-1233.6	27.30	<.001

Table S4: Test of the main effect of Trial Type for critical trials (Polysemy vs. Homonymy) in Experiment 1.

implausible conclusions by design. The best-fitting model (output in Table S5) for filler trials included:

- Fixed effects:

- Trial Type (Valid Filler vs. Invalid Filler; treatment-coded with Invalid Filler as reference)
- Syllogism Schema (Darii, Datisi, Dimatis, Disamis; sum-coded)
- Trial Type $\times$ Schema interaction
- Response time (RT; centered and scaled)
- Trial number (ItemNum; centered and scaled)
- Premise 1 plausibility (stim.premiseJudgment_P1; centered and scaled)
- Premise 2 plausibility (stim.premiseJudgment_P2; centered and scaled)

- Random effects:

- By-subject random intercepts and random slopes for Trial Type
- By-item (middle term) random intercepts

Full model specification as an R formula:

```
Response ~ TrialType * SyllogismSimple + RT + ItemNum +
PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 + TrialType | SubjectID) + (1 | TargetWord)
```

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit (Table S6).

The random slopes model provided significantly better fit and was retained for subsequent comparisons. Next, we tested whether the interaction between Trial Type and Syllogism Schema improved model fit (Table S7).

The interaction term significantly improved fit and was retained. Finally, we confirmed the main effect of Trial Type by comparing the full model with Trial Type to a nested model without it (Table S8).

S3 Detailed Analyses: Experiment 2

Experiment 2 was designed to match conclusion plausibility across Polysemy and Homonymy conditions, eliminating belief bias as a potential confound. To identify suitable arguments, we wrote multiple versions of arguments for each middle term (using different "end terms"¹⁷ S and

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-0.49 [-0.88, -0.11]	-2.49	.013	0.61 [0.42, 0.9]
Valid Filler	3.71 [3.03, 4.38]	10.77	< .001	40.73 [20.75, 79.96]
Darii	0.2 [-0.05, 0.45]	1.56	.118	1.22 [0.95, 1.57]
Datisi	0.31 [0.07, 0.54]	2.51	.012	1.36 [1.07, 1.72]
Dimatis	-0.21 [-0.46, 0.05]	-1.60	.109	0.81 [0.63, 1.05]
RT	-0.04 [-0.14, 0.07]	-0.65	.517	0.97 [0.87, 1.07]
Trial Number	-0.02 [-0.12, 0.08]	-0.38	.707	0.98 [0.89, 1.09]
Premise1 Strength	0.09 [-0.16, 0.34]	0.71	.478	1.09 [0.85, 1.4]
Premise2 Strength	0.04 [-0.16, 0.24]	0.38	.701	1.04 [0.85, 1.27]
Valid Filler x Darii	0.31 [-0.12, 0.74]	1.40	.162	1.36 [0.88, 2.1]
Valid Filler x Datisi	-0.6 [-0.98, -0.21]	-3.06	.002	0.55 [0.38, 0.81]
Valid Filler x Dimatis	-0.07 [-0.46, 0.32]	-0.36	.718	0.93 [0.63, 1.37]

Table S5: Output of the best-fitting model for filler trials (Structurally Valid vs. Invalid) in Experiment 1. Bracketed values are 95% confidence intervals. $N = 3,168$ trials from 99 participants.

Model	npar	AIC	BIC	logLik	χ^2	p
Random Intercepts Only	14	2878.4	2963.2	-1425.2	–	–
Random Slopes	16	2789.6	2886.5	-1378.8	92.80	<.001

Table S6: Comparison of random effects structures for filler trials (Structurally Valid vs. Invalid) in Experiment 1.

P), and ran an independent norming study to assess the plausibility (mean truth judgment) of their conclusions if presented in isolation (Section S5.3). Balancing of conclusions was performed by a simple algorithmic selection procedure: we wrote code to generate all possible combinations of polysemy and homonymy arguments containing a single instance of each middle term. We filtered this set such that polysemy and homonymy argument sets' mean conclusion truth judgment differed by less than 0.001 and manually selected a single set of arguments in order to maximize naturalness. One homonymous argument was excluded due to a typographical error discovered after data collection, resulting in 11 homonymous arguments and 12 polysemous arguments for this experiment. This did not affect the balanced nature of mean conclusion plausibility for analyzed data which was 0.31 for both polysemous and homonymous arguments in Experiment 2.

As with Experiment 1, we fit two sets of mixed-effects logistic regression models for Experiment 2: one comparing Valid and Invalid filler arguments, and one comparing Polysemous and Homonymous critical arguments.

S3.1 Critical Trials: Polysemy vs. Homonymy

The best-fitting model (output in Table S9) for critical trials included:

Model	npar	AIC	BIC	logLik	χ^2	p
Main Effects Only	13	2794.2	2873.0	-1384.1	–	–
With Interaction	16	2789.6	2886.5	-1378.8	10.67	.014

Table S7: Test of Trial Type $\times$ Schema interaction for filler trials (Structurally Valid vs. Invalid) in Experiment 1.

Model	npar	AIC	BIC	logLik	χ^2	<i>p</i>
Without Trial Type	12	2853.6	2926.3	-1414.8	–	–
With Trial Type + Interaction	16	2789.6	2886.5	-1378.8	72.05	<.001

Table S8: Test of the main effect of Trial Type (and interaction) for filler trials (Structurally Valid vs. Invalid) in Experiment 1.

- Fixed effects:

- Trial Type (Polysemy vs. Homonymy ; treatment-coded with Homonymy as reference)
- Response time (RT ; centered and scaled)
- Trial number (ItemNum ; centered and scaled)
- Conclusion plausibility (stim.ConcTruth ; centered and scaled)
- Premise 1 plausibility (stim.premiseJudgment_P1 ; centered and scaled)
- Premise 2 plausibility (stim.premiseJudgment_P2 ; centered and scaled)

- Random effects:

- By-subject random intercepts
- By-item (middle term) random intercepts

Syllogism schema was not included as a predictor in Experiment 2 based on the design in which each middle terms were randomly assigned to exactly one schema (rather than crossed with schemas as in Experiment 1) to facilitate conclusion plausibility matching. This is additionally in line with results from Experiment 1 showing no significant interaction schema and trial type in predicting participants' validity responses.

Full model specification as an R formula:

```
Response ~ TrialType + RT + ItemNum + ConcTruth +
PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 | SubjectID) + (1 | TargetWord)
```

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-0.11 [-0.78, 0.55]	-0.33	.739	0.89 [0.46, 1.73]
Polysemy	1.03 [0.48, 1.58]	3.68	< .001	2.81 [1.62, 4.87]
RT	-0.08 [-0.34, 0.18]	-0.61	.541	0.92 [0.71, 1.19]
Trial Number	0.01 [-0.23, 0.25]	0.06	.954	1.01 [0.79, 1.28]
Conclusion Strength	0.82 [0.51, 1.13]	5.23	< .001	2.27 [1.67, 3.08]
Premise1 Strength	-0.74 [-1.22, -0.26]	-3.03	.002	0.48 [0.3, 0.77]
Premise2 Strength	-0.6 [-1.08, -0.12]	-2.45	.014	0.55 [0.34, 0.89]

Table S9: Output of the best-fitting model for critical trials (Polysemy and Homonymy) in Experiment 2. Bracketed values are 95% confidence intervals. *N* = 644 trials from 28 participants.

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit over a model with only random intercepts (Table S10).

The random slopes model did not significantly improve fit. We therefore retained the simpler random intercepts model. We then confirmed the main effect of Trial Type by comparing the model with Trial Type to a nested model without it (Table S11).

Model	npar	AIC	BIC	logLik	χ^2	<i>p</i>
Random Intercepts Only	9	692.8	733.0	-337.4	–	–
Random Slopes	11	694.6	743.7	-336.3	2.24	.327

Table S10: Comparison of random effects structures for critical trials (Polysemy vs. Homonymy) in Experiment 2.

Model	npar	AIC	BIC	logLik	χ^2	<i>p</i>
Without Trial Type	8	701.7	737.4	-342.8	–	–
With Trial Type	9	692.8	733.0	-337.4	10.86	<.001

Table S11: Test of the main effect of Trial Type for critical trials (Polysemy vs. Homonymy) in Experiment 2.

S3.2 Filler Trials: Structurally Valid vs. Invalid

As with Experiment 1, conclusion plausibility was excluded from filler models because it was not statistically separable from structural validity. In filler trials, structurally invalid arguments systematically featured implausible conclusions by design. The best-fitting model (output in Table S12) for filler trials included:

- Fixed effects:
 - TrialType(Valid Filler vs. Invalid Filler; treatment-coded with Invalid Filler as reference)
 - Response time (RT; centered and scaled)
 - Trial number (ItemNum; centered and scaled)
 - Premise 1 plausibility (stim.premiseJudgment_P1; centered and scaled)
 - Premise 2 plausibility (stim.premiseJudgment_P2; centered and scaled)
- Random effects:
 - By-subject random intercepts and random slopes for Trial Type
 - By-item (middle term) random intercepts

Full model specification as an R formula:

```
Response ~ TrialType + RT + ItemNum +
PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 + TrialType | SubjectID) + (1 | TargetWord)
```

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit (Table S13).

The random slopes model provided significantly better fit and was retained. We then confirmed the main effect of Trial Type (Table S14).

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-0.19 [-0.82, 0.44]	-0.60	.55	0.82 [0.44, 1.55]
Valid Filler	2.86 [1.95, 3.77]	6.14	< .001	17.45 [7.01, 43.45]
RT	-0.1 [-0.32, 0.12]	-0.89	.375	0.91 [0.73, 1.13]
Trial Number	-0.12 [-0.35, 0.11]	-1.03	.301	0.89 [0.71, 1.11]
Premise1 Strength	-0.08 [-0.47, 0.31]	-0.41	.684	0.92 [0.63, 1.36]
Premise2 Strength	0.03 [-0.32, 0.38]	0.18	.86	1.03 [0.73, 1.47]

Table S12: Output of the best-fitting model for filler trials (Structurally Valid vs. Invalid) in Experiment 2. Bracketed values are 95% confidence intervals. $N = 896$ trials from 28 participants.

Model	npar	AIC	BIC	logLik	χ^2	p
Random Intercepts Only	8	846.8	885.1	-415.4	–	–
Random Slopes	10	829.1	877.1	-404.6	21.64	<.001

Table S13: Comparison of random effects structures for filler trials (Structurally Valid vs. Invalid) in Experiment 2.

S4 Detailed Analyses: Experiment 3

We fit two sets of mixed-effects logistic regression models for Experiment 3: one comparing Valid and Invalid filler arguments, and one comparing arguments with semantically Related and Unrelated word pairs. Experiment 3 replaced the polysemous and homonymous middle terms with semantically controlled word pairs to test whether semantic relatedness alone (independent of lexical identity) could support deductive inference and thus drive the Uncommon Sense Effect. Here we report the full model specifications and selection procedures.

S4.1 Critical Trials: Related vs. Unrelated Word Pairs

The best-fitting model (output in Table S15) for critical trials included:

- Fixed effects:
 - Trial Type (`Related` vs. `Unrelated` ; treatment-coded with `Unrelated` as reference)
 - Response time (`RT` ; centered and scaled)
 - Trial number (`ItemNum` ; centered and scaled)
 - Conclusion plausibility (`stim.ConcTruth` ; centered and scaled)
 - Premise 1 plausibility (`stim.premiseJudgment_P1` ; centered and scaled)
 - Premise 2 plausibility (`stim.premiseJudgment_P2` ; centered and scaled)
- Random effects:

Model	npar	AIC	BIC	logLik	χ^2	p
Without Trial Type	9	857.2	900.4	-419.6	–	–
With Trial Type	10	829.1	877.1	-404.6	30.08	<.001

Table S14: Test of the main effect of Trial Type for filler trials (Structurally Valid vs. Invalid) in Experiment 2.

- By-subject random intercepts
- By-item (word pair) random intercepts

Syllogism schema was not included as a predictor in Experiment 3 based on the design (as with Experiment 2) in which each middle word pair was randomly assigned to exactly one schema (rather than crossed with schemas as in Experiment 1).

Full model specification as an R formula:

```
Response ~ TrialType + RT + ItemNum + ConcTruth +
PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 | SubjectID) + (1 | TargetWord)
```

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-2.59 [-3.22, -1.95]	-7.99	< .001	0.08 [0.04, 0.14]
Related	0.85 [0.26, 1.44]	2.84	.005	2.34 [1.3, 4.22]
RT	0.03 [-0.19, 0.25]	0.26	.797	1.03 [0.82, 1.29]
Trial Number	-0.24 [-0.52, 0.03]	-1.72	.086	0.78 [0.59, 1.04]
Conclusion Strength	0.68 [0.36, 0.99]	4.24	< .001	1.97 [1.44, 2.69]
Premise1 Strength	0.08 [-0.26, 0.42]	0.45	.654	1.08 [0.77, 1.52]
Premise2 Strength	0.04 [-0.28, 0.36]	0.24	.81	1.04 [0.76, 1.43]

Table S15: Output of the best-fitting model for critical trials (Related vs. Unrelated word pairs) in Experiment 3. Bracketed values are 95% confidence intervals. $N = 744$ trials from 31 participants.

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit over a model with only random intercepts (Table S16).

Model	npar	AIC	BIC	logLik	χ^2	p
Random Intercepts Only	9	594.7	636.2	-288.4	–	–
Random Slopes	11	593.9	644.7	-286.0	4.78	.092

Table S16: Comparison of random effects structures for critical trials (Related vs. Unrelated) in Experiment 3.

The random slopes model did not significantly improve fit. We therefore retained the simpler random intercepts model. We then observed a main effect of Trial Type by comparing the model with Trial Type to a nested model without it (Table S17).

Model	npar	AIC	BIC	logLik	χ^2	p
Without Trial Type	8	599.6	636.5	-291.8	–	–
With Trial Type	9	594.7	636.2	-288.4	6.93	.008

Table S17: Test of the main effect of Trial Type for critical trials (Related vs. Unrelated) in Experiment 3.

We include this analysis as consistent with our pre-registration, however, as we note in the main text this apparent main effect of relatedness appears to be driven by the belief bias: Related arguments in our stimuli had higher average conclusion plausibility ($M = 0.279, SD = 0.226$)

than Unrelated arguments ($M = 0.139, SD = 0.151$), a gap of 0.14 which almost exactly maps onto the validity judgments. Most importantly, participants do not actually judge these word-pair arguments to be valid at a rate higher than the apparent plausibility of their conclusions presented in isolation. Rather, the Related and Unrelated items in Experiment 3 were endorsed as valid *less frequently* than their isolated conclusions; a change in acceptance of -0.034 and -0.036 , respectively. Contrary to an account that attributes USE to semantic relatedness alone, when adjusted for conclusion plausibility, a simple linear regression revealed no difference in item-level argument boost between Related and Unrelated conditions ($t(22) = 0.025, p = .980$), with both slightly below zero.

S4.2 Filler Trials: Structurally Valid vs. Invalid

As with Experiments 1 and 2, conclusion plausibility was excluded from filler models because it was not statistically separable from structural validity. In filler trials, structurally invalid arguments systematically featured implausible conclusions by design. The best-fitting model (output in Table S18) for filler trials included:

- Fixed effects:
 - Trial Type (Valid Filler vs. Invalid Filler; treatment-coded with Invalid Filler as reference)
 - Response time (RT; centered and scaled)
 - Trial number (ItemNum; centered and scaled)
 - Premise 1 plausibility (stim.premiseJudgment_P1; centered and scaled)
 - Premise 2 plausibility (stim.premiseJudgment_P2; centered and scaled)
- Random effects:
 - By-subject random intercepts and random slopes for Trial Type
 - By-item (middle term) random intercepts

Full model specification as an R formula:

```
Response ~ TrialType + RT + ItemNum +
PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 + TrialType | SubjectID) + (1 | TargetWord)
```

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-0.8 [-1.31, -0.28]	-3.03	.002	0.45 [0.27, 0.76]
Valid Filler	3.33 [2.55, 4.11]	8.39	< .001	27.94 [12.83, 60.82]
RT	-0.17 [-0.36, 0.02]	-1.71	.087	0.85 [0.7, 1.02]
Trial Number	0.04 [-0.17, 0.25]	0.37	.714	1.04 [0.84, 1.29]
Premise1 Strength	-0.19 [-0.52, 0.14]	-1.12	.261	0.83 [0.6, 1.15]
Premise2 Strength	-0.1 [-0.38, 0.19]	-0.66	.511	0.91 [0.68, 1.21]

Table S18: Output of the best-fitting model for filler trials (Structurally Valid vs. Invalid) in Experiment 3. Bracketed values are 95% confidence intervals. $N = 992$ trials from 31 participants.

Model	npar	AIC	BIC	logLik	χ^2	p
Random Intercepts Only	8	946.8	986.0	-465.4	–	–
Random Slopes	10	938.7	987.7	-459.3	12.11	.002

Table S19: Comparison of random effects structures for filler trials (Structurally Valid vs. Invalid) in Experiment 3.

We conducted a series of nested model comparisons to determine the optimal model structure. We first tested whether including by-subject random slopes for Trial Type improved model fit (Table S19).

The random slopes model provided significantly better fit and was retained. We then confirmed the main effect of Trial Type (Table S20).

Model	npar	AIC	BIC	logLik	χ^2	p
Without Trial Type	9	984.0	1028.1	-483.0	–	–
With Trial Type	10	938.7	987.7	-459.3	47.29	<.001

Table S20: Test of the main effect of Trial Type for filler trials (Structurally Valid vs. Invalid) in Experiment 3.

S5 Norming studies for premise and conclusion plausibility

We conducted six independent norming studies to assess the plausibility of the individual statements (conclusions and premises) which comprise arguments used in Experiments 1-3 in the main text as well as Supplementary Experiment 1 reported in the SI. As with the other experiments reported here, participants completed the study in their web browser after being recruited online via Prolific, following the same constraints as all reported studies here (see Methods). All norming studies were preregistered: April 7th, 2025 (norming studies 1 and 2); November 12th, 2024 (norming study 3); February 24th, 2025 (norming study 4); and May 14th, 2025 (norming studies 5 and 6). In all of these studies, participants were shown individual statements in isolation—i.e., presented without the argumentative context of the main studies—and were instructed to click a button marked “True” if the statement seemed true and a button marked “False” if the statement seemed false. After viewing examples of true and false statements, participants provided judgments, in sequence, on each statement in their assigned stimulus list. These norming responses were used as the basis for deriving stimulus-level plausibility values—the mean responses across all participants who judged each isolated statement—which were used as the covariates “Premise1 Strength,” “Premise2 Strength,” and “Conclusion Strength” in the mixed-effects models for experiments in the main text. Below, we provide details specific to each norming study.

S5.1 Norming study 1: conclusion plausibility for Main Experiment 1

This norming study was conducted on the plausibility of each argument’s conclusion in Experiment 1. Stimuli comprised four experimental lists, each with conclusions from 56 arguments (224 conclusions total). Statements were presented in two randomized orders thus generating eight between-subjects conditions. Participants were excluded if their mean response time across all trials was more than two standard deviations above or below the sample mean, they failed to

respond on at least 10% of trials, or they judged to be true more than 70% of the (intended to be false) statements used in Experiment 1's structurally invalid filler arguments. We recruited 82 subjects, retaining 79 after applying these registered exclusion criteria. The mean judgment for each statement was used as the "Conclusion Strength" covariate in the mixed-effects models for Experiment 1.

S5.2 Norming study 2: premise plausibility for Main Experiment 1

This norming study was conducted on the plausibility of each premise in Experiment 1. Stimuli comprised four experimental lists, each with both premises from 56 arguments (224 arguments total; 448 premises total). Statements from each list were presented in two randomized orders thus generating eight between-subjects conditions consisting of 112 premises (56 total arguments $\times$ 2 premises per argument). We planned to exclude participants if their mean response time across all trials was more than two standard deviations above or below the sample mean or if they failed to respond on at least 10% of trials. No participants met these criteria so all 80 recruited participants were retained. The mean judgment for each statement was used as the "Premise1 Strength" and "Premise2 Strength" covariates in the mixed-effects models for Experiment 1.

S5.3 Norming study 3: conclusion plausibility for Main Experiment 2

This norming study was conducted on the plausibility of conclusions for Experiment 2. As described in the Methods for Experiment 2, we initially generated 88 candidate arguments to allow downsampling for exact matching of conclusion plausibility across polysemous and homonymous conditions. This norming study gathered plausibility judgments for all 88 conclusions from this superset (31 from polysemous arguments, 25 from homonymous arguments, 16 from valid fillers, and 16 from invalid fillers). We planned to exclude participants if their mean response time across all trials was more than three standard deviations above or below the sample mean or if they failed to respond on at least 10% of trials. No participants met these criteria so all 40 recruited participants were retained. The mean judgment for each statement was used as the "Conclusion Strength" covariate in the mixed-effects models for Experiment 2 and Supplementary Experiment 1.

Additionally, using these plausibility norms, we selected a balanced subset of 56 arguments for Experiment 2 as follows. To generate a list of stimuli with balanced conclusion plausibility, we began by evenly dividing the 12 selected polysemes and 12 selected homonyms between Dimatis, Disamis, Datisi and Darii so that each syllogistic form was associated with three polysemes and three homonyms. For each of the word-argument pairs, we constructed either two or three argument candidates — with the end goal of selecting a single candidate per word-argument pair to include in the main study. This procedure yielded a total set of 88 arguments (whose conclusions we normed above): 56 ambiguous target arguments (25 polysemy and 31 homonymy) along with 16 valid fillers and 16 invalid fillers evenly divided among the four argument schemas. Our goal was to generate a conclusion-matched list of 56 arguments with the same proportion of middle term types as the lists used in Experiment 1: 12 polysemy arguments, 12 homonymy arguments, 16 valid fillers, and 16 invalid fillers. To do so, we first generated all possible combinations of the variants for each word-argument pair (22,057,920 in total). Among those, we identified the polysemy/homonymy combinations which had exactly equal mean truth judgments (830 argument lists) and manually selected the final stimulus set for use in Experiment 2 from among them.

S5.4 Norming study 4: premise plausibility for Main Experiment 2

This norming study was conducted on the plausibility of premises for Experiment 2. Following the downsampling procedure using the conclusion plausibility norms from Section S5.3 we normed the 112 premises (each argument containing two premises) from the selected 56 arguments. Participants were assigned to one of two between-subjects conditions, judging the same statements in different orders. In the first order, premises were presented in a sequential order that matched the full argument: that is, for each argument, its first premise was immediately followed by its second premise. This allowed us to gather judgments on premises in the same context as they appeared in Experiment 2. However, we also sought to gather truth judgments of second premises outside of the immediate context of the corresponding first premises. To do so, we also created a second, pseudorandomized order that met the following constraints: (i) no two premises from the same argument appeared adjacent to one another; and (ii) for half the arguments, the first premise appeared first in the list and for the other half the second premise appeared first. We recruited forty participants, evenly divided between the two conditions. As preregistered, we excluded participants if their mean response time across all trials was more than two standard deviations above or below the sample mean or if they failed to respond on at least 10% of trials. This removed only a single participants, thus we retained data from 39 participants for analysis. The mean judgment for each statement was used as the “Premise1 Strength” and “Premise2 Strength” covariates in the mixed-effects models for Experiment 2 and Supplementary Experiment 1.

S5.5 Norming study 5: conclusion plausibility for Main Experiment 3

This norming study was conducted on the plausibility of each argument’s conclusion in Experiment 3. Stimuli comprised 56 isolated conclusions (12 from semantically related word pair arguments, 12 from semantically unrelated word pair arguments, 16 from valid fillers, and 16 from invalid fillers). Statements were presented in one of two randomized orders as a between-subjects condition. We recruited 29 participants, retaining 26 after applying the following exclusion criteria: Participants were excluded if their mean response time across all trials was more than two standard deviations above or below the sample mean, they failed to respond on at least 10% of trials, or they judged to be true more than 50% of the (intended to be false) statements used in Experiment 3’s structurally invalid filler arguments. The mean judgment for each statement was used as the “Conclusion Strength” covariate in the mixed-effects models for Experiment 3.

S5.6 Norming study 6: premise plausibility for Main Experiment 3

This norming study was conducted on the plausibility of each argument’s premises in Experiment 3. Stimuli comprised 112 premises (two from each of the 56 arguments). Premises were presented sequentially by argument (i.e., each argument’s first premise was immediately followed by its second premise) in two pseudorandomized orders that differed only in the ordering of premise pairs within the full list. We planned to exclude participants if their mean response time across all trials was more than two standard deviations above or below the sample mean or if they failed to respond on at least 10% of trials. No participants met these criteria so we retained data from all 30 recruited participants. The mean judgment for each statement was used as the “Premise1 Strength” and “Premise2 Strength” covariates in the mixed-effects models for Experiment 3.

S6 Supplementary Experiment 1: Persistence of USE under Cognitive load

Experiments 1 and 2 in the main text established the Uncommon Sense Effect under freely-paced (self-guided) reading conditions, allowing participants unlimited time to process arguments and render judgments without distraction. One potential explanation for USE is that it is a task artifact: the asymmetry in accepting just equivocal polysemous arguments arises since participants have the opportunity to deliberate (with full cognitive resources at hand) or even realize something about the lexical ambiguity in question. If USE arises from such deliberative reasoning (rather than the default mode of processing arguments), then it should be eliminated when cognitive resources are constrained; though see Section S8 for various analyses which are inconsistent with such a “deliberative” task-effect interpretation. Alternatively, if USE reflects automatic inference over underspecified representations operating as a default process, it should persist even under cognitive load. To distinguish these possibilities, we conducted a supplementary experiment in which participants evaluated the same arguments as Experiment 2 while under various forms of cognitive load. We employed two conceptually distinct load manipulations: time pressure (limiting reading and deliberation time), and concurrent memory load (occupying working memory resources during judgment), allowing us to test whether USE is robust across different types of cognitive constraint.

S6.1 Methods

A total of 118 participants were recruited via Prolific and compensated at approximately \$15 per hour, following the same recruitment criteria as Experiments 1-3. This study was preregistered on December 10th, 2024, and as preregistered, participants were excluded if their mean response time across all trials was more than two standard deviations above or below the sample mean, or they failed to respond on at least 10% of trials.

Participants were randomly assigned to one of four between-subjects cognitive load conditions: 750ms time pressure ($N = 29$), 1500ms time pressure ($N = 30$), 3-digit memorization ($N = 29$), or 5-digit memorization ($N = 30$). Within each load condition, participants were further assigned to one of two randomized stimulus orders, the same stimulus lists as used in Experiment 2.

The experiment employed a mixed design with cognitive load condition (750ms time pressure, 1500ms time pressure, 3-digit memorization, 5-digit memorization) crossed with stimulus list/order as a between-subjects factors and Trial type (Polysemy, Homonymy, Valid Filler, Invalid Filler) as a within-subjects factor. All other design features and all stimuli were otherwise the same as Experiment 2 — this includes retaining the intentionally matched conclusion plausibility and incidentally matched premise plausibility of stimuli across Homonymy and Polysemy conditions. The only change was the cognitive load manipulations imposed. For participants in the 750ms and 1500ms conditions, each statement within an argument was presented for a fixed duration before advancing automatically. Specifically, the first premise (P1) appeared on screen for either 750ms or 1500ms (depending on condition), then disappeared. The second premise (P2) then appeared for the same duration, followed by the conclusion (C) for the same duration. After all three statements had been presented sequentially and automatically removed from the screen, participants were asked to judge whether the conclusion followed from the premises by selecting “yes” or “no.” For participants in the 3-digit and 5-digit conditions, participants were tasked with completing a concurrent memory task for each argument trial. At the beginning of each trial, participants were shown either three or five randomly generated digits (depending on condition) presented simultaneously on screen for 2500ms. The argument then appeared in the same self-controlled

sequential presentation format as Experiments 1-3 in the main text: participants pressed a key to advance from the first premise (P1) to the second (P2) and then to the conclusion (C). After viewing all three statements, participants judged whether the conclusion followed from the premises by selecting “yes” or “no.” Finally, participants were prompted to recall the digit sequence by typing their response in a free-response text box. All other aspects of the procedure matched those of Experiment 2.

S6.2 Results

In line with the studies in the main text, participants reliably discriminated structurally valid (87% accepted) from invalid filler (40% accepted) arguments involving unambiguous terms. This difference was confirmed by nested-model comparison: we fit a mixed-effects logistic regression model to filler trial data with trial type (Valid Filler vs. Invalid Filler), load condition (750ms, 1500ms, 3-digit, 5-digit; sum-coded), and their interaction as fixed effects, along with the same covariates as Experiment 2 (response time, trial number, premise plausibilities). The model included by-subject random intercepts and random slopes for trial type, and by-item random intercepts. The full model including trial type provided a significantly better fit than the nested model which dropped the main effect of trial type ($\chi^2(4) = 78.66, p < .001$) or the trial type $\times$ load condition interaction ($\chi^2(3) = 9.2, p = 0.027$). The significant interaction with trial type indicates that individual load conditions taxed performance to different degrees. This aligns with intuition with the less constraining 1500ms condition showing the strongest discrimination of structural validity (argument acceptance for valid: 89%, invalid: 36%) while the 750ms condition showed somewhat reduced discrimination (valid: 83%, invalid: 48%) and the digit-memorization conditions in between these two (3-digit: valid 88%, invalid 40%; 5-digit: valid 86%, invalid 37%). Despite this variation, participants in all conditions maintained clear sensitivity to structural validity, confirming that they remained engaged with the task and capable of performing it even under cognitive load.

Aside from generally weakened performance on filler trials, the expected effects of the varying difficulty of the load manipulation can be seen in digit memorization accuracy. Participants in the 3-digit condition recalled the digit sequence correctly substantially more often ($M = 0.837, SD = 0.185$) than participants in the more challenging 5-digit condition ($M = 0.714, SD = 0.251$).

Having established that participants maintained sensitivity to structural validity under cognitive load and that the memory manipulations successfully imposed additional processing demands, we turn to the critical comparison between polysemous and homonymous arguments. Replicating the findings from Experiments 1 and 2, participants under cognitive load judged polysemous arguments to be valid more often than structurally matched homonymous arguments (56% vs. 44%). Notably, the Uncommon Sense Effect persisted across all four cognitive load conditions (see Figure S1) with polysemous arguments accepted more often than homonymous arguments under both time pressure conditions (750ms: 57% vs. 52%; 1500ms: 51% vs. 33%), as well as both digit memorization conditions (3-digit: 54% vs. 44%; 5-digit: 60% vs. 48%).

The significance of this difference and its stability across conditions was confirmed by nested-model comparison. We fit a mixed-effects logistic regression model to critical trial data following the same structure as Experiment 2, with load condition (750ms, 1500ms, 3-digit, 5-digit; sum-coded) added as an additional between-subjects factor. The model included trial type (Polysemy vs. Homonymy; treatment-coded with Homonymy as reference), load condition, their interaction, and the same covariates as Experiment 2 (response time, trial number, conclusion plausibility, premise plausibilities) as fixed effects. Random effects included by-subject random intercepts and random slopes for trial type, and by-item (middle term) random intercepts. Full model specification as an R formula below:

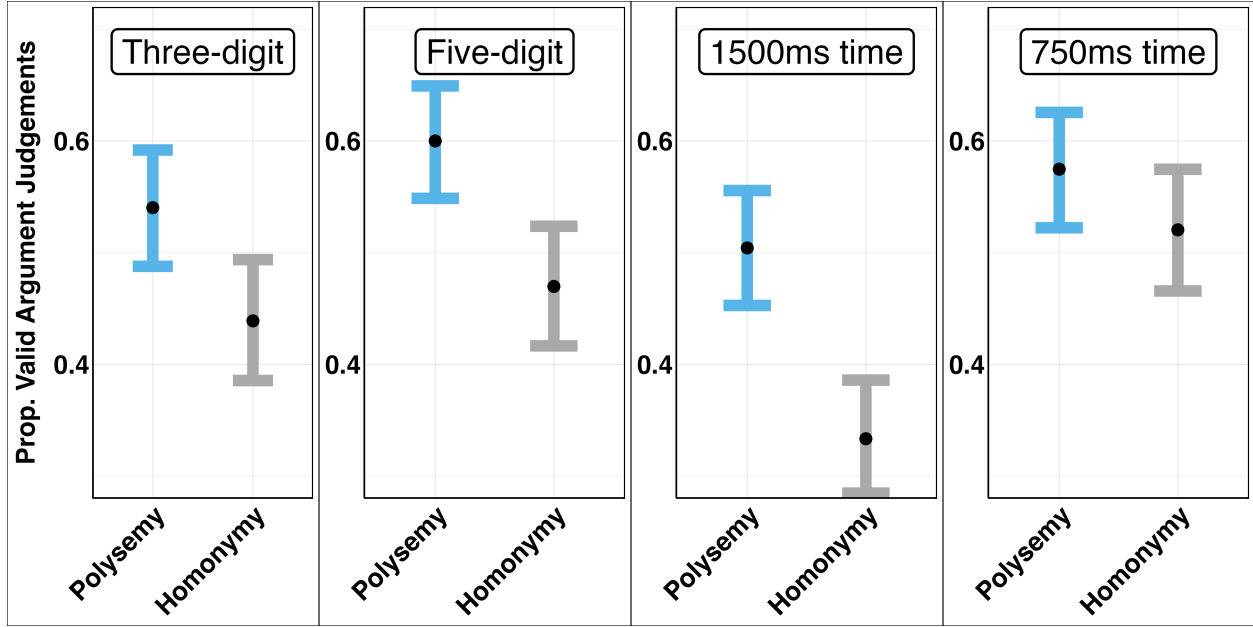


Figure S1: Proportion of arguments judged valid in the within-subjects polysemous and homonymous conditions in Supplementary Experiment 1. Each panel shows one of the four between-subjects load conditions: limited proposition display time (750ms or 1500ms) or digit-memorization (3-digit or 5-digit). Points show mean trial-level acceptance rates; error bars indicate 95% confidence intervals.

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Response ~ TrialType * loadtime + RT + ItemNum +
ConcTruth + PremiseJudgmentP1 + PremiseJudgmentP2 +
(1 + TrialType | SubjectID) + (1 | TargetWord)

```

A model including the main effect of trial type (see Table S21) provided a significantly better fit than a model which drops trial type ($\chi^2(1) = 5.86, p = 0.015$). Importantly, including the trial type $\times$ load condition interaction did not significantly improve model fit ($\chi^2(3) = 4.71, p = 0.195$), indicating that USE did not systematically vary across cognitive load manipulations—the best-fitting model therefore included main effects of trial type and load condition without their interaction. These results replicate the Uncommon Sense Effect, demonstrating that the asymmetry between the judged validity of polysemous and homonymous arguments does not depend on the availability of cognitive resources for strategic reinterpretation or effortful deliberation. The persistence of USE across time pressure and concurrent memory load conditions, instead, supports the interpretation that the effect reflects natural inference over underspecified representations.

S7 Premise Plausibility

A potential alternative explanation for the Uncommon Sense Effect is that participants rely on a simple plausibility-based heuristic rather than performing logical inference over underspecified representations. Under this account, participants might accept arguments when one or both premises sound plausible, regardless of logical structure or equivocation. If polysemous arguments systematically featured more plausible premises than homonymous arguments, this could explain the observed asymmetry without appealing to underspecified representations. We conducted

Fixed effect	Coefficient	z	p	Odds Ratio
(Intercept)	-0.34 [-0.76, 0.08]	-1.60	.11	0.71 [0.47, 1.08]
Polysemy	0.62 [0.14, 1.1]	2.54	.011	1.86 [1.15, 3]
1500ms Load	-0.36 [-0.74, 0.02]	-1.87	.062	0.7 [0.48, 1.02]
750ms Load	0.11 [-0.28, 0.5]	0.53	.594	1.11 [0.75, 1.64]
Five-Digit Load	0.29 [-0.09, 0.67]	1.49	.136	1.34 [0.91, 1.95]
RT	-0.06 [-0.19, 0.07]	-0.92	.36	0.94 [0.82, 1.07]
Trial Number	-0.05 [-0.17, 0.07]	-0.85	.395	0.95 [0.85, 1.07]
Conclusion Strength	0.63 [0.38, 0.87]	5.03	< .001	1.87 [1.47, 2.39]
Premise1 Strength	-0.4 [-0.78, -0.01]	-2.02	.043	0.67 [0.46, 0.99]
Premise2 Strength	-0.34 [-0.74, 0.05]	-1.71	.087	0.71 [0.48, 1.05]

Table S21: Output of the best-fitting model for critical trials (Polysemy and Homonymy) in Supplementary Experiment 1. Bracketed values are 95% confidence intervals. $N = 2,713$ trials from 118 participants.

multiple analyses to test this alternative explanation across Experiments 1 and 2, with converging lines of evidence that premise plausibility cannot explain USE.

The data do not support such a premise-plausibility heuristic theory: this view would rely on polysemous arguments having systematically more plausible premises than homonymous arguments, but our independent norming data (SI, Sections S5.2 and S5.4) revealed no such differences. The mean plausibility of each argument's first premise (P1) did not differ across conditions in Experiment 1 (polysemy: $M = 0.59$, $SD = 0.30$; homonymy: $M = 0.63$, $SD = 0.31$; $t(94) = 0.75$, $p = .454$, 95% CI $[-0.08, 0.17]$). Likewise, the mean plausibility of the second premise (P2) did not differ (polysemy: $M = 0.67$, $SD = 0.28$; homonymy: $M = 0.58$, $SD = 0.35$; $t(89.9) = -1.35$, $p = .179$, 95% CI $[-0.22, 0.04]$). Even if participants were sensitive just to arguments for which both premises were independently plausible, combined premise plausibility—computed simply the mean of P1 and P2 for each argument—also did not differ (polysemy: $M = 0.63$, $SD = 0.12$; homonymy: $M = 0.61$, $SD = 0.08$; $t(83.59) = -1.03$, $p = .306$, 95% CI $[-0.06, 0.020]$).

The same holds for Experiment 2, which was explicitly designed to match conclusion plausibility across critical conditions. Premise 1 plausibility did not differ (polysemy: $M = 0.60$, $SD = 0.29$; homonymy: $M = 0.63$, $SD = 0.31$; $t(20.5) = 0.21$, $p = .833$, 95% CI $[-0.24, 0.29]$), nor did Premise 2 plausibility (polysemy: $M = 0.66$, $SD = 0.33$; homonymy: $M = 0.65$, $SD = 0.30$; $t(21.0) = -0.11$, $p = .917$, 95% CI $[-0.29, 0.26]$). Combined premise plausibility was also not significantly different (polysemy: $M = 0.63$, $SD = 0.11$; homonymy: $M = 0.64$, $SD = 0.07$; $t(18.6) = 0.17$, $p = .865$, 95% CI $[-0.07, 0.09]$). Together then, stimuli in Experiment 2 were controlled across conditions for the plausibility of both premises P1 and P2 (treated independently or taken together) as well as conclusions, strongly undermining an explanation of USE that appeals to premise plausibility.

Nevertheless, for completeness, we also confirmed whether premise plausibility (separate from middle term condition) had an effect on argument judgments. Counter to a premise-plausibility heuristic view—under which premise plausibility should positively predict acceptance—we see either no correlation or a negative correlation between premise plausibility and argument acceptance. This is apparent in the full regression outputs (Tables S1, S5, S9, S12). In Experiment 1, Premise 1 plausibility showed no relationship with acceptance of critical trials ($OR = 0.96$, 95% CI $[0.70, 1.32]$, $p = .805$), while Premise 2 plausibility exhibited a significant *negative* relationship ($OR = 0.65$, 95% CI $[0.52, 0.82]$, $p < .001$): higher premise plausibility was associated with *lower* acceptance rates. In Experiment 2, both premises showed significant negative relationships with

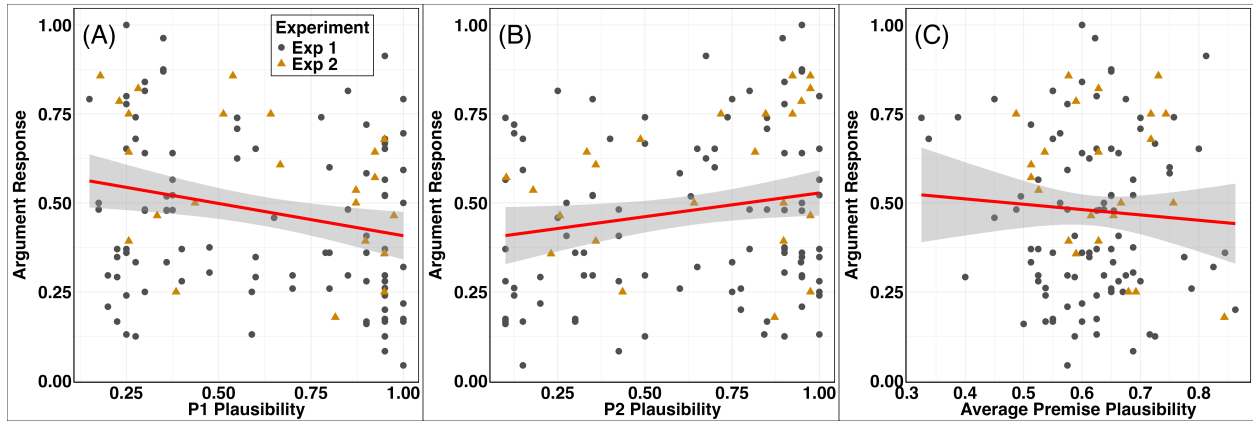


Figure S2: Scatterplot showing the null or weakly negative relationship between premise plausibility and argument acceptance in critical trials (polysemy and homonymy). Left panel (A) shows the relationship between each argument's first premise (P1) plausibility and argument acceptance rate; middle panel (B) shows the relationship between each argument's second premise (P2) plausibility and argument acceptance rate; right panel (C) shows the relationship between each argument's combined premise plausibility (mean of P1 and P2) and argument acceptance rate. Dark grey circles represent arguments from Experiment 1; gold triangles represent arguments from Experiment 2. Red lines show linear regression fits with 95% confidence regions (light grey shaded areas). The absence of any reliable positive correlations contradicts predictions of a premise plausibility-based heuristic account.

acceptance of critical trials (P1: $OR = 0.48$, 95% CI $[0.30, 0.77]$, $p = .002$; P2: $OR = 0.55$, 95% CI $[0.34, 0.89]$, $p = .014$). For filler trials, premise plausibility showed no systematic relationship with acceptance in either Experiment 1 (P1: $OR = 1.09$, 95% CI $[0.85, 1.40]$, $p = .478$; P2: $OR = 1.04$, 95% CI $[0.85, 1.27]$, $p = .701$) or Experiment 2 (P1: $OR = 0.92$, 95% CI $[0.63, 1.36]$, $p = .684$; P2: $OR = 1.03$, 95% CI $[0.73, 1.47]$, $p = .860$).

Premise plausibility cannot serve as an explanation for the Uncommon Sense Effect: there were no differences in premise plausibility between conditions in either Experiment 1 or Experiment 2. Even if there had been a latent different in premise strength across conditions, premise plausibility showed either a null or negative relationships with argument acceptance. We visualize this null / weakly negative relationship in Figure S2.

S8 Pragmatic Coercion

Our explanation of the Uncommon Sense Effect draws on the nature of words' underlying representations, rather than the plausibility of an argument's conclusion. We controlled for, and found no evidence to support, this latter "belief bias" explanation in Experiments 1 and 2 in the main text. However, one might nevertheless object that the plausibility of a conclusion presented in isolation is not directly comparable to the plausibility of a conclusion in the context of a complete argument. This is because the interpretation of a sentence is a function not only of its literal semantic content but also of its presentation in a wider discourse context. The additional context provided by the premises may have suggested alternative, more plausible, interpretations of the conclusion. We term this view "conclusion coercion." If conclusions are pragmatically coerced in a full argumentative context, our conclusion norming studies may not have adequately ruled out

a different kind of belief bias explanation of USE.

In order for conclusion coercion to explain USE, there must be an interaction between middle term condition (polysemy vs. homonymy) and conclusion coercibility, with polysemous argument conclusions gaining more plausibility from the additional argumentative context relative to homonymous argument conclusions. For example, consider the arguments in Table S22. On this alternative, when participants read the polysemy argument, they interpret “dinners” in the event sense in (P1) but in the food sense in (P2). However, when they read the conclusion, they reinterpret “Some foods are black tie” as meaning either some food events are black tie or some foods are black tie foods. Participants would consequently judge the arguments to be valid on the basis of the plausibility of the non-literal, coerced conclusion. In contrast, when participants read the homonymy argument, there is no plausible interpretation of either the S term (“animals”) or P term (“made for softball”) that makes the conclusion plausible, so a non-literal reinterpretation is unavailable. Thus, participants cannot judge the homonymy arguments to be valid on the basis of any plausible non-literal, coerced conclusion.

Polyseme (Conclusion Coercion)	Homonym (no coercion)
(P1) Some dinners _{EVENT} are black tie	(P1) Some bats _{TOOL} are made for softball
(P2) All dinners _{FOOD} are foods	(P2) All bats _{ANIMAL} are animals
(C) Some foods are black tie [foods]	(C) Some animals are made for softball

Table S22: Hypothetical interpretation of an argument under an alternative “conclusion coercion” theory. Under this view, participants coerce a non-literal interpretation of the polysemy conclusion (e.g., interpreting “black tie” as modifying an implicit event), making it seem plausible despite equivocation. No such coercion is available for the homonymy argument.

S8.1 Analysis of Reading and Response Times

Analysis shows that this pragmatic coercion explanation is inconsistent with the data. If judgments of validity for polysemy arguments but not homonymy arguments involve reinterpretation of the literal meaning of conclusions, then total response time (reading and judgment latency) for polysemy arguments should be slower than for homonymy arguments, reflecting the additional cognitive effort required for pragmatic reinterpretation.

Contrary to this prediction, we find no difference in response times between polysemous and homonymous arguments. In Experiment 1, mean reading and response times for polysemous arguments ($M = 10,980$ ms, $SD = 5,391$ ms) did not differ from those for homonymous arguments ($M = 11,108$ ms, $SD = 5,594$ ms; Welch’s $t(2248) = -0.55$, $p = .579$). The same pattern held in Experiment 2: reading and response for polysemous arguments ($M = 10,384$ ms, $SD = 4,621$ ms) did not differ from those for homonymous arguments ($M = 10,172$ ms, $SD = 4,336$ ms; Welch’s $t(624) = 0.59$, $p = .555$). The absence of any response time difference between conditions is inconsistent with an account on which participants engage in pragmatic reinterpretation specifically for polysemous arguments. This suggests that acceptance of polysemous arguments reflects a more automatic, default inferential process operating over underspecified representations, rather than a task specific reinterpretation.

One may object that judgments for both polysemy and homonymy arguments involve an attempted reinterpretation of the literal meaning of the arguments, but such a reinterpretation is only successful in the polysemy case. However, if both sorts of arguments involve an attempted reinterpretation, it is not obvious that both of these attempts should, on average, take the same

amount of time; that is, it is not clear why a successful reinterpretation (in the polysemy cases) should take the same amount of time as an unsuccessful reinterpretation (in the homonymy cases).

S8.2 Null effect of trial number

More decisively than reading+response time patterns, the conclusion coercion explanation makes a clear prediction about trial order effects that is disconfirmed by the data. Previous research has shown that over time, listeners may make strategic adjustments in how they interpret linguistic utterances⁹³. If participants in our study are pragmatically reinterpreting polysemy arguments, they would be more likely to do so towards the end of the experimental session, after having encountered multiple polysemous arguments and perhaps having noticed that the arguments shown may be somewhat “clever” or atypical. The pragmatic coercion account thus predicts an interaction between trial number and middle term condition: the later a polysemy argument is seen, the more likely it should be judged valid (if participants increasingly engage in pragmatic coercion), relative to judgments of validity for homonymous arguments.

The data strongly contradict this prediction. In Experiment 1, there was no main effect of trial number on validity judgments ($\chi^2(1) = 2.90, p = .088$), and critically, no interaction between polysemy condition and trial number ($\chi^2(1) = 0.01, p = .919$). This pattern replicated in Experiment 2: no main effect of trial number ($\chi^2(1) = 0.00, p = .955$) and no interaction with trial type ($\chi^2(1) = 0.00, p = .989$).

Figure S3 visualizes this stability over time in Experiment 1 (where there is the largest number of trials, allowing for reliable estimates of argument acceptance as a function of trial number). The acceptance rate for polysemous arguments remains consistently higher than for homonymous arguments across the entire experimental session, with no evidence of increasing divergence or convergence between conditions. If participants were gradually learning to apply pragmatic coercion strategies specifically to polysemous arguments, we would expect the gap between conditions to widen over time. The absence of this pattern is inconsistent with a pragmatic coercion account and instead suggests that the Uncommon Sense Effect reflects a stable, default inferential process that operates over underspecified representations from the first trial onward.

S8.3 Premise ordering

Perhaps instead of just coercing the conclusion, participants also coerce the second premise when provided with the additional context of the first premise. This explanation could be called “premise-and-conclusion coercion.” More concretely, this alternative would suggest that when participants read the polysemy argument, they interpret “dinner” in the plausible event sense in (P1), but then they stick with that same sense in (P2)—see Table S23. As a result, they coerce the non-middle term “foods” in (P2) to make that premise more plausible, reading “All dinners are food” to mean something along the lines of all dinner *events* are food events. Then, when participants read the conclusion, they read the second instance of the non-middle term from (P2), “foods” in the same, coerced sense, food events. In contrast, no comparable coercion is available for the homonymy arguments. Notably, this coercion story allows the reinterpreted arguments to be strictly valid (at the denotational level).

This premise-and-conclusion coercion explanation inherits the problems of the conclusion coercion explanation and makes an additional, disconfirmed prediction. Note that the Darii and Datisi schemas form a minimal pair: both contain a universal “All M are P” as their first premise (P1), and both conclusions (C) “Some S are P” are the same. The two schemas differ only in that

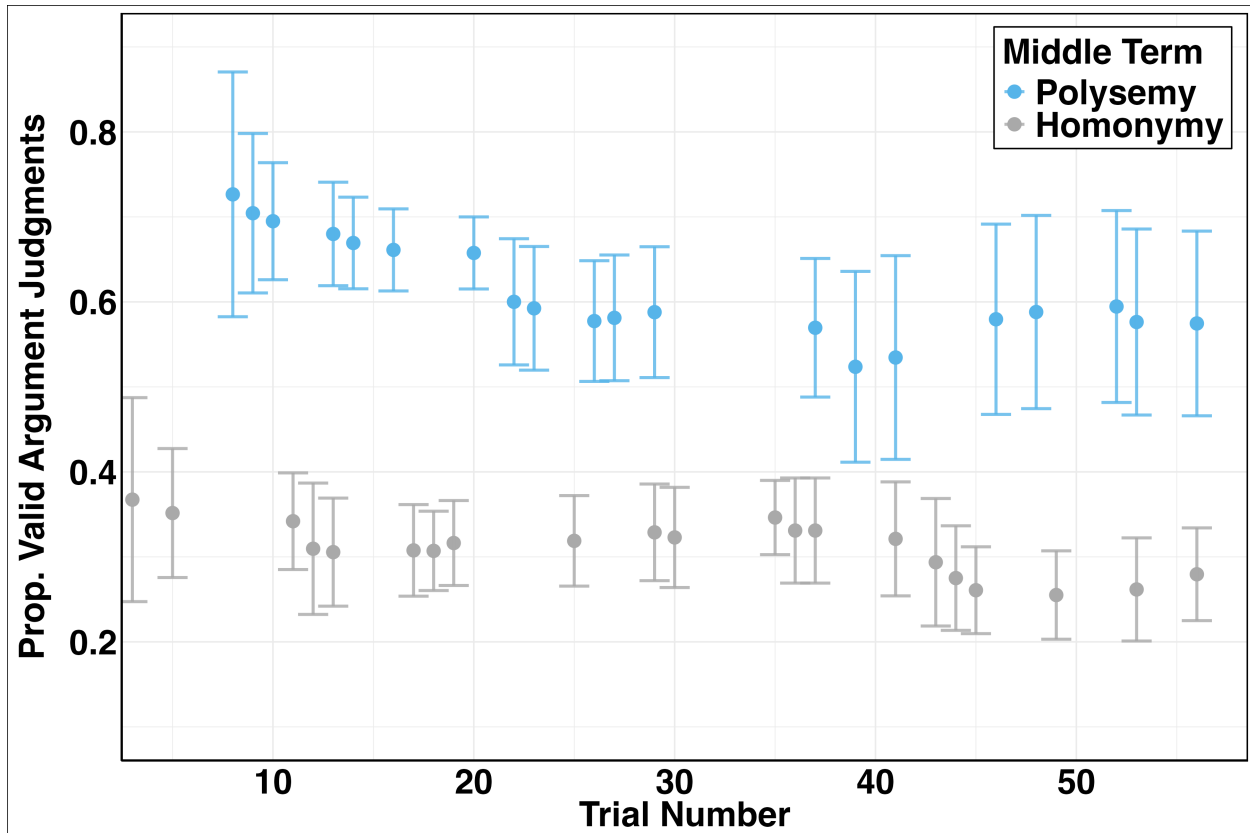


Figure S3: Mean rates of validity judgment of Polysemous (blue) and Homonymous (grey) arguments in Experiment 1 as a function of trial number. Points represent rolling means computed over an 8-trial window; error bars show 95% confidence intervals. The Uncommon Sense Effect remains stable throughout the experimental session, with no interaction between trial number and trial type ($\chi^2(1) = 0.01, p = .919$).

for Datisi, the middle term comes first in (P2), while in Darii, the middle term comes second in (P2). See Table S24.

We can use this minimal pair to draw out a prediction concerning the proportion of validity judgments. If participants are maintaining the (P1) sense of the polyseme when it occurs in (P2), and consequently coercing the S term to fit this sense, then coercion (and thus validity judgments) should be more likely in the case where one reads the second occurrence of the polyseme before reading the S term (Datisi), compared to when one reads the second occurrence of the polyseme after reading the S term (Darii). For if one reads the S term first, there is not yet any pragmatic pressure to reinterpret it non-literally (as in Darii). For instance, if one reads “Some sandwiches are lunches,” there is no pressure to reinterpret “sandwiches” as sandwich-eating events, since any pragmatic reason to do so is given by the subsequent occurrence of “lunch,” which is by hypothesis, read in the event sense. However, there would be pressure if one reads “lunch” in the event sense, before the S term (as in Datisi). This coercion story thus predicts that validity judgments for polysemous Datisi arguments should be more frequent than those for polysemous Darii arguments.

Again, however, the data do not support such a premise-and-conclusion coercion explanation as its predictions are not confirmed. We originally tested whether Datisi arguments were accepted

Polyseme (Premise-and-Conclusion Coercion)	Homonym (no coercion)
(P1) Some dinners _{EVENT} are black tie (P2) All dinners _{EVENT} are food [event]s (C) Some food [event]s are black tie	(P1) Some bats _{TOOL} are made for softball (P2) All bats _{ANIMAL} are animals (C) Some animals are made for softball

Table S23: Hypothetical interpretation of an argument under an alternative “premise-and-conclusion coercion” theory.

	Schema	Premise-and-Conclusion Coercion
Datisti	(P1) All M are P (P2) Some M are S (C) Some S are P	(P1) All lunches _{EVENT} are breaks (P2) Some lunches _{EVENT} are soup [eating event]s (C) Some soup [eating event]s are breaks
Darii	(P1) All M are P (P2) Some S are M (C) Some S are P	(P1) All lunches _{EVENT} are breaks (P2) Some sandwich [eating event]s are lunches _{EVENT} (C) Some sandwich [eating event]s are breaks

Table S24: Hypothetical interpretation of arguments under an alternative “Premise-and-Conclusion coercion” theory. The Darri and Datisti schemas form a minimal pair where the relative order of the M and S vary in the second premise.

more often than Darri arguments by including syllogism schema as a predictor in our Experiment 1 models (see Section S2). If the premise-and-conclusion coercion account were correct, we should observe a positive interaction between Trial Type (Polysemy vs. Homonymy) and Schema (Datisti vs. Darri), with Datisti showing a larger polysemy advantage than Darri. Instead, we found no such interaction ($\chi^2(3) = 2.65, p = .449$; see Table S3).

To examine the data more intuitively, we extracted just the polysemous Darri and Datisti arguments from Experiments 1 and 2. In Experiment 1, acceptance rates for polysemous arguments did not differ significantly between Darri (65.1%) and Datisti (66.9%) schemas ($\beta = -0.05, SE = 0.17, z = -0.26, p = .794$). This null result replicated in Experiment 2: Darri (76.8%) versus Datisti (80.0%) ($\beta = -0.16, SE = 0.39, z = -0.40, p = .690$). Therefore, we find no support for the premise-and-conclusion coercion explanation.

In light of these arguments, USE does not appear to be driven by either a belief bias based on pragmatic reinterpretations of polysemous arguments’ conclusions in light of their premises, or validity judgments based on pragmatic reinterpretations of polysemous arguments’ premises and conclusions.

S9 Filler performance

We have argued USE shows that individuals can perform rule-based inferences over underspecified representations. As already reported in Experiments 1 and 2, one piece of evidence that supports the claim that individuals are completing our task using an inferential strategy is that they reliably discriminate valid from invalid fillers in the correct way. While this result is consistent with sensitivity to validity, it is not decisive. For instance, participants’ discrimination could also be explained by a latent difference in conclusion plausibility for valid and invalid filler items. As a matter of fact, valid and invalid conclusions were not matched in our experiments, and valid fillers did tend to have plausible conclusions while invalid fillers had broadly implausible ones. Thus, individuals merely attending to conclusion plausibility could explain participants’ apparent ability

to discriminate structurally valid from invalid fillers (though this could not explain the Uncommon Sense Effect in Experiment 2, as polysemous and homonymous arguments were matched to have strictly equivalent conclusion plausibility across conditions).

In light of this concern, we marshal a more decisive piece of evidence. Our experiment uses middle-term condition as a within-subject factor; that is, each participant was tested on arguments containing equivocal polysemes and homonyms, as well as on structurally valid and invalid fillers. Since each participant completed arguments in every middle term condition, we are able to analyze both the magnitude of each individual's USE (by comparing their responses on polysemy and homonymy trials), and the degree to which each individual correctly judges valid and invalid fillers. Examining the relationship between participant-level filler trial performance and USE magnitude can provide further evidence that USE is driven by inference. If participants are using rule-based inference and USE is driven by it, then both filler performance and the size of the USE should reflect a participant's underlying capacity to apply inference rules during the experiment. Thus, these variables should covary across individuals: the better an individual performs on the filler trials, the larger the magnitude of their USE. By contrast, if individuals were merely attending to conclusion plausibility or using some other non-inferential strategy, we would predict no such relationship.

The former prediction is borne out in analyses of Experiments 1 and 2: individuals' performance on filler trials positively correlates with the magnitude of their attested USE. We classified participants into three groups based on their filler trial performance: "Ceiling" performers (who correctly judged $\geq 85\%$ of filler trials), "Good" performers (who correctly judged 70-85% of filler trials), and "Bad" performers (who correctly judged $< 70\%$ of filler trials). We then computed each participant's USE score (acceptance rate for polysemous arguments minus acceptance rate for homonymous arguments) and examined how USE magnitude varied across performance groups.

Figure S4 shows a clear monotonic relationship between filler performance and USE magnitude in both experiments. In Experiment 1, Ceiling performers showed the largest USE (mean difference = 0.432, 95% CI [0.353, 0.511], $n = 22$), followed by Good performers (mean difference = 0.312, 95% CI [0.248, 0.377], $n = 44$), and Bad performers (mean difference = 0.215, 95% CI [0.123, 0.307], $n = 33$). This pattern replicated in Experiment 2: Ceiling performers (mean difference = 0.267, 95% CI [0.112, 0.421], $n = 5$), Good performers (mean difference = 0.219, 95% CI [0.056, 0.382], $n = 10$), and Bad performers (mean difference = 0.092, 95% CI [0.015, 0.169], $n = 13$).

Critically, even participants who performed poorly on filler trials showed a positive USE, indicating that the effect is not simply an artifact of high-performing reasoners. However, the magnitude of USE systematically increased with filler performance, demonstrating that participants who were better at discriminating structurally valid from invalid arguments also showed a stronger tendency to accept polysemous over homonymous arguments. This pattern is precisely what would be predicted if USE reflects the application of the same inferential cognitive machinery that supports performance on filler trials: namely, rule-based deductive inference operating over underspecified representations.

S10 Representation of Homonymy

In presenting the logic of our experiment, we assumed that polysemes contain a single orthographic representation, a single underspecified representation, and multiple denotation-determining representations (Figure 1, left). This assumption is in line with standard accounts of polysemes' underlying representations, especially the regular polysemes we examine (³⁶⁻³⁸; cf. ^{71,73}). We also assumed that homonyms contain a single orthographic representation and multiple under-

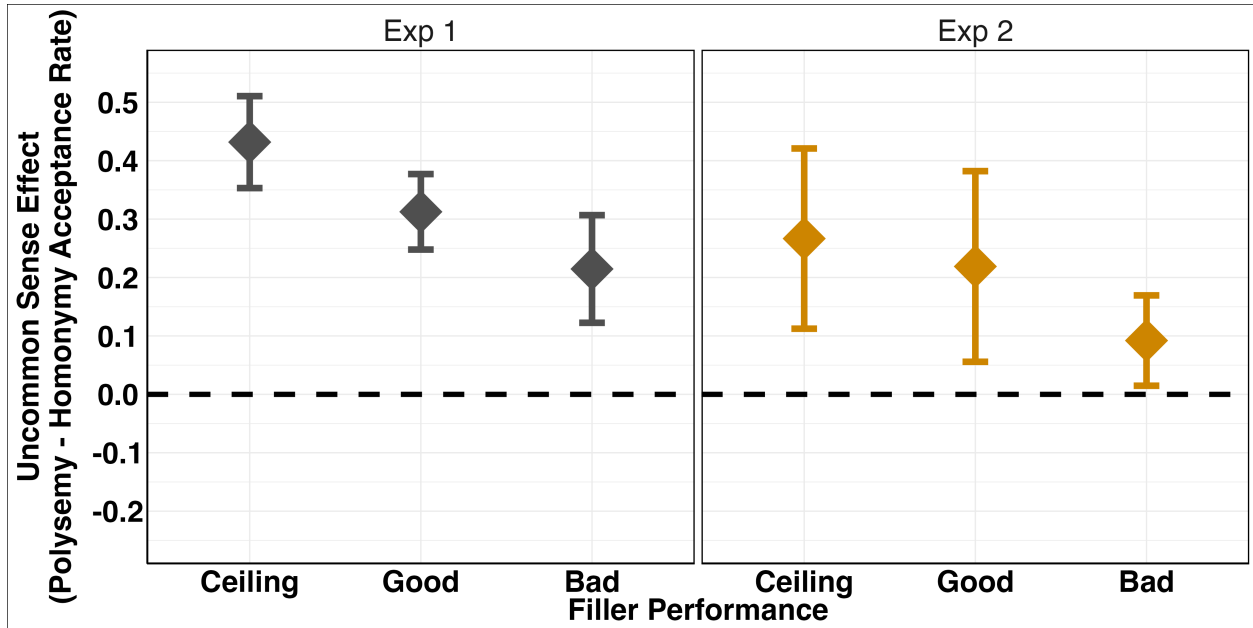


Figure S4: Magnitude of the Uncommon Sense Effect (acceptance rate for polysemous arguments minus acceptance rate for homonymous arguments) as a function of filler trial performance in Experiments 1 and 2. Participants were classified into three groups based on their accuracy on filler trials: Ceiling ($\geq 85\%$ correct), Good (70-85% correct), and Bad ($< 70\%$ correct). Error bars represent 95% confidence intervals. Better filler performance is associated with larger USE magnitude, suggesting that both reflect participants' underlying application of rule-based deductive inference.

specified representations, where each underspecified representation maps to its own denotation-determining representation (Figure 1, right). However, another possible view is that homonyms contain a single orthographic representation that maps directly to multiple denotation-determining representations, without an intermediate underspecified representation (Figure S5, right). We assumed the former view of homonyms not only because it is more representationally uniform, but also because it can explain the previously-observed fact that homonyms can contain polysemous meanings^{40;45}. For instance, while “bank” is homonymous between riparian and financial meanings, the financial meaning is polysemous between physical and institutional senses (compare: “he walked into the bank to deposit a check” and “the bank fired its president”). This observation is difficult to make sense of on a view where homonyms directly map orthographic representations to denotation-determining representations (as in Figure 1, right) but is easily accommodated by the view according to which homonyms contain underspecified representations, which themselves can branch to multiple denotation-determining representations.

While we find this assumption about the representations of homonyms plausible, we note that one could reinterpret our experiment on the alternative view of homonym representation. If homonyms map directly from an orthographic representation to multiple denotation-determining representations, without underspecified intermediaries, it is still true that the arguments we present to participants would share the same orthographic form across premises (P1, P2) and would therefore be valid if evaluated at the orthographic level. Likewise, both polysemes and homonyms would equivocate at the denotational level: the premises employ different senses (e.g., dinner as food vs. event) or unrelated meanings (e.g., bat as an animal vs. tool), rendering the arguments denotationally invalid. Finally, polysemes but not homonyms would share an under-

Level of representation	Polysemy		Homonymy	
Orthographic		(P1) Some dinners are P (P2) All dinners are S (C) Some S are P		(P1) Some bats are P (P2) All bats are S (C) Some S are P
Underspecified		(P1) Some dinners are P (P2) All dinners are S (C) Some S are P		
Denotational	 	(1) Some dinners are P (2) All dinners are S (3) Some S are P	 	(P1) Some bats are P (P2) All bats are S (C) Some S are P

Figure S5: Illustration of how lexical ambiguity may instantiate valid deductive inference schemas at some, but not all, levels of representation, even on an alternative view of homonym representation that does not involve underspecified representations. Both polysemous (left) and homonymous (right) arguments share the same orthographic form across premises (P1, P2) and are therefore valid if evaluated at the orthographic level. Both kinds of words equivocate at the denotational level: the premises employ different senses (e.g., dinner as food vs. event) or unrelated meanings (e.g., bat as an animal vs. tool), rendering the arguments denotationally invalid. Critically, polysemes but not homonyms share an underspecified representation which is neutral between senses, rendering the polysemy argument valid at this intermediate level.

specified representation which is neutral between senses, rendering the polysemy argument valid at this intermediate level. Thus, USE could still be taken to suggest that participants can reason over underspecified representations. However, on this view, the natural interpretation of participants' rejection of homonym arguments would be that they evaluate the argument instantiated at the denotation-determining level. This interpretation is distinct from that given in the main text according to which both polysemes and homonyms are evaluated according to the argument instantiated at the underspecified level.

Further work could examine homonyms with ostensibly polysemous senses, like "bank", and look for an USE between arguments equivocating on their senses (e.g., All banks are near water; Some companies are banks; Some companies are near water) versus those equivocating on their meanings (e.g., All banks contain vaults; Some companies are banks; Some companies contain vaults). To the extent that participants show USE on such arguments, this would be evidence for the claim that homonyms, too, have underspecified representations that can be used in reasoning. This possibility raises the likelihood that all words – even unambiguous ones – contain an underspecified representation. It is just that for many homonyms and unambiguous words, such an underspecified representation is difficult to tease apart from its underlying denotation-determining representation because their one-to-one mapping.

S11 Complete Stimuli

Table S25: Full argument set from Experiment 1

Middle Term	Condition	Syllogism	Argument
book	Polysemy	Darii	All books are bound Some exposés are books Some exposés are bound
breakfast	Polysemy	Darii	All breakfasts are under three hours Some smoothies are breakfasts Some smoothies are under three hours
city	Polysemy	Darii	All cities are bureaucracies Some graffitied places are cities Some graffitied places are bureaucracies
college	Polysemy	Darii	All colleges grant degrees Some historic buildings are colleges Some historic buildings grant degrees
company	Polysemy	Darii	All companies are buildings Some tech giants are companies Some tech giants are buildings
dinner	Polysemy	Darii	All dinners happen after four o'clock Some lasagnas are dinners Some lasagnas happen after four o'clock
letter	Polysemy	Darii	All letters have stamps Some rejections are letters Some rejections have stamps
lunch	Polysemy	Darii	All lunches are breaks Some sandwiches are lunches Some sandwiches are breaks
magazines	Polysemy	Darii	All magazines are recyclable Some media corporations are magazines Some media corporations are recyclable
novel	Polysemy	Darii	All novels are flammable Some debuts are novels Some debuts are flammable
picnic	Polysemy	Darii	All picnics happen in nature Some salads are picnics Some salads happen in nature
school	Polysemy	Darii	All schools have classrooms Some employers are schools Some employers have classrooms
book	Polysemy	Datisi	All books are printed Some books are romances Some romances are printed
breakfast	Polysemy	Datisi	All breakfasts are under three hours Some breakfasts are smoothies Some smoothies are under three hours
city	Polysemy	Datisi	All cities are bureaucracies Some cities are graffitied places Some graffitied places are bureaucracies
college	Polysemy	Datisi	All colleges grant degrees Some colleges are historic buildings Some historic buildings grant degrees

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
company	Polysemy	Datisi	All companies are buildings Some companies are hierarchies Some hierarchies are buildings
dinner	Polysemy	Datisi	All dinners happen after four o'clock Some dinners are meatloafs Some meatloafs happen after four o'clock
letter	Polysemy	Datisi	All letters have stamps Some letters are rejections Some rejections have stamps
lunch	Polysemy	Datisi	All lunches are breaks Some lunches are soups Some soups are breaks
magazines	Polysemy	Datisi	All magazines are recyclable Some magazines are media corporations Some media corporations are recyclable
novel	Polysemy	Datisi	All novels are flammable Some novels are debuts Some debuts are flammable
picnic	Polysemy	Datisi	All picnics happen in nature Some picnics are salads Some salads happen in nature
school	Polysemy	Datisi	All schools have hallways Some schools are employers Some employers have hallways
book	Polysemy	Dimatis	Some doorstops are books All books are pieces of intellectual property Some pieces of intellectual property are doorstops
breakfast	Polysemy	Dimatis	Some meetings are breakfasts All breakfasts are nutrients Some nutrients are meetings
city	Polysemy	Dimatis	Some bureaucracies are cities All cities are graffitied places Some graffitied places are bureaucracies
college	Polysemy	Dimatis	Some funding sources are colleges All colleges are places filled with desks Some places filled with desks are funding sources
company	Polysemy	Dimatis	Some storefronts are companies All companies are hierarchies Some hierarchies are storefronts
dinner	Polysemy	Dimatis	Some black tie events are dinners All dinners are foods Some foods are black tie events
letter	Polysemy	Dimatis	Some papers are letters All letters are communications Some communications are papers
lunch	Polysemy	Dimatis	Some breaks are lunches All lunches are foods Some foods are breaks
magazines	Polysemy	Dimatis	Some recyclables are magazines All magazines are subscriptions Some subscriptions are recyclables

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
novel	Polysemy	Dimatis	Some flammable objects are novels All novels are fictions Some fictions are flammable objects
picnic	Polysemy	Dimatis	Some summertime activities are picnics All picnics are prepared foods Some prepared foods are summertime activities
school	Polysemy	Dimatis	Some rectangular buildings are schools All schools are employers Some employers are rectangular buildings
book	Polysemy	Disamis	Some books are doorstops All books are pieces of intellectual property Some pieces of intellectual property are doorstops
breakfast	Polysemy	Disamis	Some breakfasts are under an hour All breakfasts are nutrients Some nutrients are under an hour
city	Polysemy	Disamis	Some cities are corrupt All cities are groups of buildings Some groups of buildings are corrupt
college	Polysemy	Disamis	Some colleges have chancellors All colleges are groups of buildings Some groups of buildings have chancellors
company	Polysemy	Disamis	Some companies are in city centers All companies are hierarchies Some hierarchies are in city centers
dinner	Polysemy	Disamis	Some dinners are black tie All dinners are foods Some foods are black tie
letter	Polysemy	Disamis	Some letters are crumpled All letters are communications Some communications are crumpled
lunch	Polysemy	Disamis	Some lunches are breaks All lunches are foods Some foods are breaks
magazines	Polysemy	Disamis	Some magazines are coffee-stained All magazines are subscriptions Some subscriptions are coffee-stained
novel	Polysemy	Disamis	Some novels weigh over a pound All novels are stories Some stories weigh over a pound
picnic	Polysemy	Disamis	Some picnics are romantic All picnics are prepared foods Some prepared foods are romantic
school	Polysemy	Disamis	Some schools are rectangular All schools are employers Some employers are rectangular
ball	Homonymy	Darii	All balls are toys Some birthday parties are balls Some senior proms are toys
bat	Homonymy	Darii	All bats share a common ancestor Some sporting goods are bats Some sporting goods share a common ancestor

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
board	Homonymy	Darii	All boards are wooden planks Some governing organizations are boards Some governing organizations are wooden planks
cabinet	Homonymy	Darii	All cabinets are political appointments Some shelving units are cabinets Some shelving units are political appointments
change	Homonymy	Darii	All change is money Some surprises are change Some surprises are money
match	Homonymy	Darii	All matches are combustible Some pay-per-view broadcasts are matches Some pay-per-view broadcasts are combustible
ring	Homonymy	Darii	All rings are sounds Some accessories are rings Some accessories are sounds
rock	Homonymy	Darii	All rock has guitars Some marble is rock Some marble has guitars
ruler	Homonymy	Darii	All rulers are standard lengths Some queens are rulers Some queens are standard lengths
star	Homonymy	Darii	All stars have autographs Some red dwarfs are stars Some red dwarfs have autographs
suit	Homonymy	Darii	All suits are dress attire Some court proceedings are suits Some court proceedings are dress attire
Wave	Homonymy	Darii	All waves are water Some gestures are waves Some gestures are water
ball	Homonymy	Datisi	All balls are toys Some balls are senior proms Some senior proms are toys
bat	Homonymy	Datisi	All bats are animals Some bats are sporting goods Some sporting goods are animals
board	Homonymy	Datisi	All boards are wooden planks Some boards are governing organizations Some governing organizations are wooden planks
cabinet	Homonymy	Datisi	All cabinets are political appointments Some cabinets are shelving units Some shelving units are political appointments
change	Homonymy	Datisi	All change is money Some change is transformation Some transformation is money
match	Homonymy	Datisi	All matches are combustible Some matches are pay-per-view broadcasts Some pay-per-view broadcasts are combustible
ring	Homonymy	Datisi	All rings are sounds Some rings are accessories Some accessories are sounds

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
rock	Homonymy	Datissi	All rock has guitars Some rock is marble Some marble has guitars
ruler	Homonymy	Datissi	All rulers are standard lengths Some rulers are queens Some queens are standard lengths
star	Homonymy	Datissi	All stars have autographs Some stars are red dwarfs Some red dwarfs have autographs
suit	Homonymy	Datissi	All suits are dress attire Some suits are court proceedings Some court proceedings are dress attire
wave	Homonymy	Datissi	All waves are water Some waves are gestures Some gestures are water
ball	Homonymy	Dimatis	Some parties are balls All balls are toys Some toys are parties
bat	Homonymy	Dimatis	Some sporting goods are bats All bats are animals Some animals are sporting goods
board	Homonymy	Dimatis	Some governing organizations are boards All boards are wooden planks Some wooden planks are governing organizations
cabinet	Homonymy	Dimatis	Some shelving units are cabinets All cabinets are political appointments Some political appointments are shelving units
change	Homonymy	Dimatis	Some transformation is change All change is money Some money is transformation
match	Homonymy	Dimatis	Some pay-per-view broadcasts are matches All matches are fire starters Some fire starters are pay-per-view broadcasts
ring	Homonymy	Dimatis	Some accessories are rings All rings are sounds Some sounds are accessories
rock	Homonymy	Dimatis	Some marble is rock All rock is music Some music is marble
ruler	Homonymy	Dimatis	Some queens are rulers All rulers are standard lengths Some standard lengths are queens
star	Homonymy	Dimatis	Some celestial objects are stars All stars are famous people Some famous people are celestial objects
suit	Homonymy	Dimatis	Some court proceedings are suits All suits are formal outfits Some formal outfits are court proceedings
Wave	Homonymy	Dimatis	Some gestures are waves All waves are liquids Some liquids are gestures

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
ball	Homonymy	Disamis	Some balls are senior proms All balls are toys
bat	Homonymy	Disamis	Some toys are senior proms Some bats are made for softball All bats are animals Some animals are made for softball
board	Homonymy	Disamis	Some boards are governing organizations All boards are wooden planks Some wooden planks are governing organizations
cabinet	Homonymy	Disamis	Some cabinets are antique furniture All cabinets are political appointments Some political appointments are antique furniture
change	Homonymy	Disamis	Some change is transformation All change is money Some money is transformation
match	Homonymy	Disamis	Some matches are pay-per-view All matches are fire starters Some fire starters are pay-per-view
ring	Homonymy	Disamis	Some rings are made of gold All rings are sounds Some sounds are made of gold
rock	Homonymy	Disamis	Some rock is marble All rock is music Some music is marble
ruler	Homonymy	Disamis	Some rulers are kings All rulers are standard lengths Some standard lengths are kings
star	Homonymy	Disamis	Some stars are red dwarfs All stars are famous people Some famous people are red dwarfs
suit	Homonymy	Disamis	Some suits have multiple defendants All suits are dress attire Some dress attire has multiple defendants
Wave	Homonymy	Disamis	Some waves are gestures All waves are liquids Some liquids are gestures
burgers	Valid Filler	Darii	All burgers have protein Some entrées are burgers Some entrées have protein
clowns	Valid Filler	Darii	All clowns can juggle Some circus performers are clowns Some circus performers can juggle
dolls	Valid Filler	Darii	All dolls are man made Some toys are dolls Some toys are man made
doughnuts	Valid Filler	Darii	All doughnuts are sugary Some pastries are doughnuts Some pastries are sugary
grapes	Valid Filler	Darii	All grapes are fruits Some snacks are grapes Some snacks are fruits

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
liars	Valid Filler	Darii	All liars are humans Some criminals are liars Some criminals are humans
mugs	Valid Filler	Darii	All mugs are cups Some ceramics are mugs Some ceramics are cups
poets	Valid Filler	Darii	All poets are writers Some teachers are poets Some teachers are writers
ponies	Valid Filler	Darii	All ponies neigh Some thoroughbreds are ponies Some thoroughbreds neigh
rifles	Valid Filler	Darii	All rifles are guns Some safety hazards are rifles Some safety hazards are guns
singers	Valid Filler	Darii	All singers are vocalists Some guitarists are singers Some guitarists are vocalists
statues	Valid Filler	Darii	All statues are sculptures Some artworks are statues Some artworks are sculptures
suitcases	Valid Filler	Darii	All suitcases are luggage Some bags are suitcases Some bags are luggage
swords	Valid Filler	Darii	All swords are weapons Some sharp objects are swords Some sharp objects are weapons
trucks	Valid Filler	Darii	All trucks are vehicles Some cars are trucks Some cars are vehicles
widows	Valid Filler	Darii	All widows are women Some parents are widows Some parents are women
burgers	Valid Filler	Datisi	All burgers have protein Some burgers are drive-thru orders Some drive-thru orders have protein
clowns	Valid Filler	Datisi	All clowns are entertainers Some clowns are mimes Some mimes are entertainers
dolls	Valid Filler	Datisi	All dolls are man made Some dolls are toys Some toys are man made
doughnuts	Valid Filler	Datisi	All doughnuts are sugary Some doughnuts are treats Some treats are sugary
grapes	Valid Filler	Datisi	All grapes are fruits Some grapes are snacks Some snacks are fruits
liars	Valid Filler	Datisi	All liars are humans Some liars are criminals Some criminals are humans

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
mugs	Valid Filler	Datisi	All mugs are cups Some mugs are ceramics Some ceramics are cups
poets	Valid Filler	Datisi	All poets are writers Some poets are teachers Some teachers are writers
ponies	Valid Filler	Datisi	All ponies neigh Some ponies are thoroughbreds Some thoroughbreds neigh
rifles	Valid Filler	Datisi	All rifles are guns Some rifles are safety hazards Some safety hazards are guns
singers	Valid Filler	Datisi	All singers are performers Some singers are men Some men are performers
statues	Valid Filler	Datisi	All statues are sculptures Some statues are bronzes Some bronzes are sculptures
suitcases	Valid Filler	Datisi	All suitcases are pieces of luggage Some suitcases are bags Some bags are pieces of luggage
swords	Valid Filler	Datisi	All swords are weapons Some swords are museum pieces Some museum pieces are weapons
trucks	Valid Filler	Datisi	All trucks are vehicles Some trucks are big rigs Some big rigs are vehicles
widows	Valid Filler	Datisi	All widows are women Some widows are parents Some parents are women
burgers	Valid Filler	Dimatis	Some entrées are burgers All burgers are sandwiches Some sandwiches are entrées
clowns	Valid Filler	Dimatis	Some jugglers are clowns All clowns are performers Some performers are jugglers
dolls	Valid Filler	Dimatis	Some toys are dolls All dolls are man made objects Some man made objects are toys
doughnuts	Valid Filler	Dimatis	Some pastries are doughnuts All doughnuts are sugary foods Some sugary foods are pastries
grapes	Valid Filler	Dimatis	Some snacks are grapes All grapes are fruits Some fruits are snacks
liars	Valid Filler	Dimatis	Some criminals are liars All liars are humans Some humans are criminals
mugs	Valid Filler	Dimatis	Some ceramics are mugs All mugs are cups Some cups are ceramics

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
poets	Valid Filler	Dimatis	Some teachers are poets All poets are writers Some writers are teachers
ponies	Valid Filler	Dimatis	Some thoroughbreds are ponies All ponies are neighing animals Some neighing animals are thoroughbreds
rifles	Valid Filler	Dimatis	Some safety hazards are rifles All rifles are guns Some guns are safety hazards
singers	Valid Filler	Dimatis	Some YouTubers are singers All singers are musicians Some musicians are YouTubers
statues	Valid Filler	Dimatis	Some bronzes are statues All statues are sculptures Some sculptures are bronzes
suitcases	Valid Filler	Dimatis	Some bags are suitcases All suitcases are pieces of luggage Some pieces of luggage are bags
swords	Valid Filler	Dimatis	Some sharp things are swords All swords are weapons Some weapons are sharp things
trucks	Valid Filler	Dimatis	Some cars are trucks All trucks are vehicles Some vehicles are cars
widows	Valid Filler	Dimatis	Some parents are widows All widows are women Some women are parents
burgers	Valid Filler	Disamis	Some burgers are overcooked All burgers are sandwiches Some sandwiches are overcooked
clowns	Valid Filler	Disamis	Some clowns are mimes All clowns are entertainers Some entertainers are mimes
dolls	Valid Filler	Disamis	Some dolls are toys All dolls are man made objects Some man made objects are toys
doughnuts	Valid Filler	Disamis	Some doughnuts are glazed All doughnuts are sugary foods Some sugary foods are glazed
grapes	Valid Filler	Disamis	Some grapes are red All grapes are fruits Some fruits are red
liars	Valid Filler	Disamis	Some liars are criminals All liars are humans Some humans are criminals
mugs	Valid Filler	Disamis	Some mugs are ceramics All mugs are cups Some cups are ceramics
poets	Valid Filler	Disamis	Some poets are teachers All poets are writers Some writers are teachers

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
ponies	Valid Filler	Disamis	Some ponies are thoroughbreds All ponies are neighing animals Some neighing animals are thoroughbreds
rifles	Valid Filler	Disamis	Some rifles are for hunting All rifles are guns Some guns are for hunting
singers	Valid Filler	Disamis	Some singers are guitarists All singers are vocalists Some vocalists are guitarists
statues	Valid Filler	Disamis	Some statues are bronzes All statues are sculptures Some sculptures are bronzes
suitcases	Valid Filler	Disamis	Some suitcases are bags All suitcases are pieces of luggage Some pieces of luggage are bags
swords	Valid Filler	Disamis	Some swords are decorative All swords are weapons Some weapons are decorative
trucks	Valid Filler	Disamis	Some trucks are big rigs All trucks are vehicles Some vehicles are big rigs
widows	Valid Filler	Disamis	Some widows are parents All widows are women Some women are parents
actors	Invalid Filler	Darii*	All male leads are actors Some women are actors Some women are male leads
antiques	Invalid Filler	Darii*	All medieval manuscripts are antiques Some pieces of jewelry are antiques Some pieces of jewelry are medieval manuscripts
colors	Invalid Filler	Darii*	Some pastels are colors All shades of neon green are colors Some shades of neon green are pastels
dances	Invalid Filler	Darii*	All ballets are dances Some mating rituals are dances Some mating rituals are ballets
dressess	Invalid Filler	Darii*	All ball gowns are dresses Some cocktail outfits are dresses Some cocktail outfits are ball gowns
funerals	Invalid Filler	Darii*	All anonymous burials are funerals Some family gatherings are funerals Some family gatherings are anonymous burials
injuries	Invalid Filler	Darii*	All broken bones are injuries Some sunburns are injuries Some sunburns are broken bones
mothers	Invalid Filler	Darii*	All grandmothers are mothers Some teens are mothers Some teens are grandmothers
peppers	Invalid Filler	Darii*	All jalapenos are peppers Some yellow vegetables are peppers Some yellow vegetables are jalapenos

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
sausages	Invalid Filler	Darii*	All chili dogs are sausages Some pizza toppings are sausages Some pizza toppings are chili dogs
shapes	Invalid Filler	Darii*	All cylinders are shapes Some road signs are shapes Some road signs are cylinders
spices	Invalid Filler	Darii*	All mustard seeds are spices Some dried roots are spices Some dried roots are mustard seeds
steaks	Invalid Filler	Darii*	All rotten ribeyes are steaks Some delicious meals are steaks Some delicious meals are rotten ribeyes
stones	Invalid Filler	Darii*	All boulders are stones Some paperweights are stones Some paperweights are boulders
thieves	Invalid Filler	Darii*	All mob bosses are thieves Some children are thieves Some children are mob bosses
toilets	Invalid Filler	Darii*	All urinals are toilets Some seats are toilets Some seats are urinals
actors	Invalid Filler	Datisi*	All male leads are actors Some actors are women Some women are male leads
antiques	Invalid Filler	Datisi*	All medieval manuscripts are antiques Some antiques are pieces of jewelry Some pieces of jewelry are medieval manuscripts
colors	Invalid Filler	Datisi*	All shades of neon green are colors Some colors are pastels Some pastels are shades of neon green
dances	Invalid Filler	Datisi*	All ballets are dances Some dances are mating rituals Some mating rituals are ballets
dressess	Invalid Filler	Datisi*	All ball gowns are dresses Some dresses are cocktail outfits Some cocktail outfits are ball gowns
funerals	Invalid Filler	Datisi*	All open-casket services are funerals Some funerals are anonymous burials Some anonymous burials are open-casket services
injuries	Invalid Filler	Datisi*	All broken bones are injuries Some injuries are sunburns Some sunburns are broken bones
mothers	Invalid Filler	Datisi*	All grandmas are mothers Some mothers are teens Some teens are grandmas
peppers	Invalid Filler	Datisi*	All jalapenos are peppers Some peppers are yellow vegetables Some yellow vegetables are jalapenos
sausages	Invalid Filler	Datisi*	All chili dogs are sausages Some sausages are salamis Some salamis are chili dogs

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
shapes	Invalid Filler	Datisi*	All road signs are shapes Some shapes are cylinders Some cylinders are road signs
spices	Invalid Filler	Datisi*	All mustard seeds are spices Some spices are dried roots Some dried roots are mustard seeds
steaks	Invalid Filler	Datisi*	All rotten ribeyes are steaks Some steaks are delicious meals Some delicious meals are rotten ribeyes
stones	Invalid Filler	Datisi*	All pebbles are stones Some stones are boulders Some boulders are pebbles
thieves	Invalid Filler	Datisi*	All mob bosses are thieves Some thieves are children Some children are mob bosses
toilets	Invalid Filler	Datisi*	All outhouses are toilets Some toilets are urinals Some urinals are outhouses
actors	Invalid Filler	Dimatis*	Some women are actors All male leads are actors Some male leads are women
antiques	Invalid Filler	Dimatis*	Some pieces of jewelry are antiques All medieval manuscripts are antiques Some medieval manuscripts are pieces of jewelry
colors	Invalid Filler	Dimatis*	Some pastels are colors All shade of neon green are colors Some shades of neon green are pastels
dances	Invalid Filler	Dimatis*	Some mating rituals are dances All ballets are dances Some ballets are mating rituals
dressses	Invalid Filler	Dimatis*	Some cocktail outfits are dresses All ball gowns are dresses Some ball gowns are cocktail outfits
funerals	Invalid Filler	Dimatis*	Some family gatherings are funerals All anonymous burials are funerals Some anonymous burials are family gatherings
injuries	Invalid Filler	Dimatis*	Some sunburns are injuries All broken bones are injuries Some broken bones are sunburns
mothers	Invalid Filler	Dimatis*	Some teens are mothers All grandmas are mothers Some grandmas are teens
peppers	Invalid Filler	Dimatis*	Some yellow vegetables are peppers All jalapenos are peppers Some jalapenos are yellow vegetables
sausages	Invalid Filler	Dimatis*	Some pizza toppings are sausages All chili dogs are sausages Some chili dogs are pizza toppings
shapes	Invalid Filler	Dimatis*	Some road signs are shapes All cylinders are shapes Some cylinders are road signs

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
spices	Invalid Filler	Dimatis*	Some dried roots are spices All mustard seeds are spices Some mustard seeds are dried roots
steaks	Invalid Filler	Dimatis*	Some delicious entrées are steaks All rotten ribeyes are steaks Some rotten ribeyes are delicious entrées
stones	Invalid Filler	Dimatis*	Some paperweights are stones All boulders are stones Some boulders are paperweights
thieves	Invalid Filler	Dimatis*	Some children are thieves All mob bosses are thieves Some mob bosses are children
toilets	Invalid Filler	Dimatis*	Some seats are toilets All urinals are toilets Some urinals are seats
actors	Invalid Filler	Disamis*	Some actors are women All male leads are actors Some male leads are women
antiques	Invalid Filler	Disamis*	Some antiques are pieces of jewelry All medieval manuscripts are antiques Some medieval manuscripts are pieces of jewelry
colors	Invalid Filler	Disamis*	Some colors are primary All shades of purple are colors Some shades of purple are primary
dances	Invalid Filler	Disamis*	Some dances are mating rituals All ballets are dances Some ballets are mating rituals
dressses	Invalid Filler	Disamis*	Some dresses are cocktail outfits All ball gowns are dresses Some ball gowns are cocktail outfits
funerals	Invalid Filler	Disamis*	Some funerals are family gatherings All anonymous burials are funerals Some anonymous burials are family gatherings
injuries	Invalid Filler	Disamis*	Some injuries are sunburns All broken bones are injuries Some broken bones are sunburns
mothers	Invalid Filler	Disamis*	Some mothers are teens All grandmas are mothers Some grandmas are teens
peppers	Invalid Filler	Disamis*	Some peppers are yellow vegetables All jalapenos are peppers Some jalapenos are yellow vegetables
sausages	Invalid Filler	Disamis*	Some sausages are chili dogs All pepperonis are sausages Some pepperonis are chili dogs
shapes	Invalid Filler	Disamis*	Some shapes are triangles All squares are shapes Some squares are triangles
spices	Invalid Filler	Disamis*	Some spices are dried roots All mustard seeds are spices Some mustard seeds are dried roots

Table S25: Full argument set from Experiment 1 (*continued*)

Middle Term	Condition	Syllogism	Argument
steaks	Invalid Filler	Disamis*	Some steaks are filet mignons All ribeyes are steaks Some ribeyes are filet mignons
stones	Invalid Filler	Disamis*	Some stones are pebbles All boulders are stones Some boulders are pebbles
thieves	Invalid Filler	Disamis*	Some thieves are children All mob bosses are thieves Some mob bosses are children
toilets	Invalid Filler	Disamis*	Some toilets are coed All urinals are toilets Some urinals are coed

Table S26: Full argument set from Experiment 2

Middle Term	Condition	Syllogism	Argument
breakfast	Polysemy	Darii	All breakfasts are under three hours Some smoothies are breakfasts Some smoothies are under three hours
novel	Polysemy	Darii	All novels are flammable Some debuts are novels Some debuts are flammable
city	Polysemy	Datisi	All cities are bureaucracies Some cities are graffitied Some bureaucracies are graffitied
college	Polysemy	Datisi	All colleges grant degrees Some colleges are historic buildings Some historic buildings grant degrees
letter	Polysemy	Datisi	All letters have stamps Some letters are rejections Some rejections have stamps
picnic	Polysemy	Datisi	All picnics happen in nature Some picnics are salads Some salads happen in nature
book	Polysemy	Dimatis	Some doorstops are books All books are intellectual property Some intellectual property is a doorstop
company	Polysemy	Dimatis	Some storefronts are companies All companies are hierarchies Some hierarchies are storefronts
lunch	Polysemy	Dimatis	Some breaks are lunches All lunches are entrées Some entrées are breaks
dinner	Polysemy	Disamis	Some dinners are black tie All dinners are foods Some foods are black tie
magazines	Polysemy	Disamis	Some magazines are coffee-stained All magazines are subscriptions Some subscriptions are coffee-stained

Table S26: Full argument set from Experiment 2 (*continued*)

Middle Term	Condition	Syllogism	Argument
school	Polysemy	Disamis	Some schools are rectangular All schools are employers Some employers are rectangular
bat	Homonymy	Darii	All bats share a common ancestor Some sporting goods are bats Some sporting goods share a common ancestor
change	Homonymy	Darii	All change is money Some surprises are change Some surprises are money
ring	Homonymy	Darii	All rings are sounds Some accessories are rings Some accessories are sounds
cabinet	Homonymy	Datisi	All cabinets are political appointments Some cabinets are antique furniture Some antique furniture are political appointments
rock	Homonymy	Datisi	All rock has guitars Some rock is marble Some marble has guitars
star	Homonymy	Datisi	All stars are famous Some stars are red dwarfs Some red dwarfs are famous
ball	Homonymy	Dimatis	Some parties are balls All balls are toys Some toys are parties
board	Homonymy	Dimatis	Some governing organizations are boards All boards are wooden planks Some wooden planks are governing organizations
match	Homonymy	Disamis	Some matches are shown on TV All matches are fire starters Some fire starters are shown on TV
ruler	Homonymy	Disamis	Some rulers are kings All rulers are standard lengths Some standard lengths are kings
suit	Homonymy	Disamis	Some suits have multiple defendants All suits are dress attire Some dress attire has multiple defendants
clowns	Valid Filler	Darii	All clowns can juggle Some circus performers are clowns Some circus performers can juggle
dolls	Valid Filler	Darii	All dolls are man made Some toys are dolls Some toys are man made
doughnuts	Valid Filler	Darii	All doughnuts are sugary Some pastries are doughnuts Some pastries are sugary
singers	Valid Filler	Darii	All singers have an ideal vocal range Some guitarists are singers Some guitarists have an ideal vocal range
mugs	Valid Filler	Datisi	All mugs are cups Some mugs are ceramics Some ceramics are cups

Table S26: Full argument set from Experiment 2 (*continued*)

Middle Term	Condition	Syllogism	Argument
poets	Valid Filler	Datisi	All poets are writers Some poets are teachers Some teachers are writers
ponies	Valid Filler	Datisi	All ponies neigh Some ponies are colts Some colts neigh
statues	Valid Filler	Datisi	All statues are sculptures Some statues are bronzes Some bronzes are sculptures
liars	Valid Filler	Dimatis	Some criminals are liars All liars are humans Some humans are criminals
suitcases	Valid Filler	Dimatis	Some bags are suitcases All suitcases are pieces of luggage Some pieces of luggage are bags
trucks	Valid Filler	Dimatis	Some cars are trucks All trucks are vehicles Some vehicles are cars
widows	Valid Filler	Dimatis	Some mothers are widows All widows are women Some women are mothers
burgers	Valid Filler	Disamis	Some burgers are overcooked All burgers are sandwiches Some sandwiches are overcooked
grapes	Valid Filler	Disamis	Some grapes are red All grapes are fruits Some fruits are red
rifles	Valid Filler	Disamis	Some rifles are for hunting All rifles are guns Some guns are for hunting
swords	Valid Filler	Disamis	Some swords are decorative All swords are weapons Some weapons are decorative
antiques	Invalid Filler	Darii*	All medieval manuscripts are antiques Some pieces of jewelry are antiques Some pieces of jewelry are medieval manuscripts
dances	Invalid Filler	Darii*	All polkas are dances Some mating rituals are dances Some mating rituals are polkas
dresses	Invalid Filler	Darii*	All ball gowns are dresses Some cocktail outfits are dresses Some cocktail outfits are ball gowns
mothers	Invalid Filler	Darii*	All grandmothers are mothers Some teens are mothers Some grandmothers are teens
actors	Invalid Filler	Datisi*	All male leads are actors Some actors are women Some women are male leads
funerals	Invalid Filler	Datisi*	All open-casket services are funerals Some funerals are cremations Some cremations are open-casket services

Table S26: Full argument set from Experiment 2 (*continued*)

Middle Term	Condition	Syllogism	Argument
sausages	Invalid Filler	Datisi*	All hot dogs are sausages Some sausages are salamis Some hot dogs are salamis
stones	Invalid Filler	Datisi*	All pebbles are stones Some stones are boulders Some boulders are pebbles
injuries	Invalid Filler	Dimatis*	Some sunburns are injuries All cases of frostbite are injuries Some cases of frostbite are sunburns
peppers	Invalid Filler	Dimatis*	Some yellow vegetables are peppers All jalapenos are peppers Some jalapenos are yellow vegetables
spices	Invalid Filler	Dimatis*	Some dried roots are spices All mustard seeds are spices Some mustard seeds are dried roots
thieves	Invalid Filler	Dimatis*	Some children are thieves All mob bosses are thieves Some mob bosses are children
colors	Invalid Filler	Disamis*	Some colors are primary All shades of purple are colors Some shades of purple are primary
shapes	Invalid Filler	Disamis*	Some shapes are triangles All squares are shapes Some squares are triangles
steaks	Invalid Filler	Disamis*	Some steaks are filet mignons All ribeyes are steaks Some ribeyes are filet mignons
toilets	Invalid Filler	Disamis*	Some toilets are coed All urinals are toilets Some urinals are coed

Table S27: Full argument set from Experiment 3

Middle Term	Condition	Syllogism	Argument
ambulance emergency	Related	Darii	All ambulances are vehicles Some wildfires are emergencies Some wildfires are vehicles
beans rice	Related	Darii	All beans are legumes Some grains are rice Some grains are legumes
waist hip	Related	Darii	All waists fit in pants Some bones are hips Some bones fit in pants
compass direction	Related	Datisi	All compasses have a needle Some directions are instructions Some instructions have a needle

Table S27: Full argument set from Experiment 3 (*continued*)

Middle Term	Condition	Syllogism	Argument
knight armor	Related	Datise	All knights are brave Some armors are decorations Some decorations are brave
sock shoe	Related	Datise	All socks are elastic Some shoes are high heels Some high heels are elastic
blackboard chalk	Related	Dimatis	Some walls are blackboards All chalks are writing tools Some writing tools are walls
throne king	Related	Dimatis	Some seats are thrones All kings are rulers Some rulers are seats
thunder lightning	Related	Dimatis	Some rumbling is thunder All lightning is bright light Some bright light is rumbling
cloak dagger	Related	Disamis	Some cloaks are velvet All daggers are knives Some knives are velvet
jaw mouth	Related	Disamis	Some jaws are bones All mouths are holes Some holes are bones
noun verb	Related	Disamis	Some nouns are proper names All verbs are actions Some actions are proper names
journal hamburger	Unrelated	Darii	All journals are made of paper Some entrées are hamburgers Some entrées are made of paper
rabbi toothbrush	Unrelated	Darii	All rabbis are Jewish Some toiletries are toothbrushes Some toiletries are Jewish
tortoise syrup	Unrelated	Darii	All tortoises are animals Some ice cream toppings are syrups Some ice cream toppings are animals
lion religion	Unrelated	Datise	All lions are felines Some religions are cults Some cults are felines
mammal weekend	Unrelated	Datise	All mammals are warm blooded Some weekends are holidays Some holidays are warm blooded
spider soda	Unrelated	Datise	All spiders have eight legs Some sodas are diet drinks Some diet drinks have eight legs

Table S27: Full argument set from Experiment 3 (*continued*)

Middle Term	Condition	Syllogism	Argument
microscope mob	Unrelated	Dimatis	Some scientific instruments are microscopes All mobs are criminal organizations Some criminal organizations are scientific instruments
sack clam	Unrelated	Dimatis	Some bags are sacks All clams are sea creatures Some sea creatures are bags
surgery hall	Unrelated	Dimatis	Some medical procedures are surgeries All halls are passageways Some passageways are medical procedures
gang feet	Unrelated	Disamis	Some gangs commit crimes All feet are body parts Some body parts commit crimes
official cereal	Unrelated	Disamis	Some officials are democratically elected All cereals are grains Some grains are democratically elected
pepperoni alcohol	Unrelated	Disamis	Some pepperonis are pork All alcohols are intoxicants Some intoxicants are pork
clowns clowns	Valid Filler	Darii	All clowns can juggle Some circus performers are clowns Some circus performers can juggle
dolls dolls	Valid Filler	Darii	All dolls are man made Some toys are dolls Some toys are man made
doughnuts doughnuts	Valid Filler	Darii	All doughnuts are sugary Some pastries are doughnuts Some pastries are sugary
singers singers	Valid Filler	Darii	All singers are vocalists Some guitarists are singers Some guitarists are vocalists
mugs mugs	Valid Filler	Datisi	All mugs are cups Some mugs are ceramics Some ceramics are cups
poets poets	Valid Filler	Datisi	All poets are writers Some poets are teachers Some teachers are writers
ponies ponies	Valid Filler	Datisi	All ponies neigh Some ponies are thoroughbreds Some thoroughbreds neigh
statues statues	Valid Filler	Datisi	All statues are sculptures Some statues are bronzes Some bronzes are sculptures

Table S27: Full argument set from Experiment 3 (*continued*)

Middle Term	Condition	Syllogism	Argument
liars liars	Valid Filler	Dimatis	Some criminals are liars All liars are humans Some humans are criminals
suitcases suitcases	Valid Filler	Dimatis	Some bags are suitcases All suitcases are pieces of luggage Some pieces of luggage are bags
trucks trucks	Valid Filler	Dimatis	Some cars are trucks All trucks are vehicles Some vehicles are cars
widows widows	Valid Filler	Dimatis	Some parents are widows All widows are women Some women are parents
burgers burgers	Valid Filler	Disamis	Some burgers are overcooked All burgers are sandwiches Some sandwiches are overcooked
grapes grapes	Valid Filler	Disamis	Some grapes are red All grapes are fruits Some fruits are red
rifles rifles	Valid Filler	Disamis	Some rifles are for hunting All rifles are guns Some guns are for hunting
swords swords	Valid Filler	Disamis	Some swords are decorative All swords are weapons Some weapons are decorative
antiques antiques	Invalid Filler	Darii*	All medieval manuscripts are antiques Some pieces of jewelry are antiques Some pieces of jewelry are medieval manuscripts
dances dances	Invalid Filler	Darii*	All ballets are dances Some mating rituals are dances Some mating rituals are ballets
dressess dressess	Invalid Filler	Darii*	All ball gowns are dressess Some cocktail outfits are dressess Some cocktail outfits are ball gowns
mothers mothers	Invalid Filler	Darii*	All grandmothers are mothers Some teens are mothers Some teens are grandmothers
actors actors	Invalid Filler	Datisi*	All male leads are actors Some actors are women Some women are male leads
funerals funerals	Invalid Filler	Datisi*	All open-casket services are funerals Some funerals are anonymous burials Some anonymous burials are open-casket services

Table S27: Full argument set from Experiment 3 (*continued*)

Middle Term	Condition	Syllogism	Argument
sausages sausages	Invalid Filler	Datisi*	All chili dogs are sausages Some sausages are salamis Some salamis are chili dogs
stones stones	Invalid Filler	Datisi*	All pebbles are stones Some stones are boulders Some boulders are pebbles
injuries injuries	Invalid Filler	Dimatis*	Some sunburns are injuries All broken bones are injuries Some broken bones are sunburns
peppers peppers	Invalid Filler	Dimatis*	Some yellow vegetables are peppers All jalapenos are peppers Some jalapenos are yellow vegetables
spices spices	Invalid Filler	Dimatis*	Some dried roots are spices All mustard seeds are spices Some mustard seeds are dried roots
thieves thieves	Invalid Filler	Dimatis*	Some children are thieves All mob bosses are thieves Some mob bosses are children
colors colors	Invalid Filler	Disamis*	Some colors are primary All shades of purple are colors Some shades of purple are primary
shapes shapes	Invalid Filler	Disamis*	Some shapes are triangles All squares are shapes Some squares are triangles
steaks steaks	Invalid Filler	Disamis*	Some steaks are filet mignons All ribeyes are steaks Some ribeyes are filet mignons
toilets toilets	Invalid Filler	Disamis*	Some toilets are coed All urinals are toilets Some urinals are coed